2-D Unstructured Compressible Navier–Stokes Solver Benchmark Report

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August 28, 2026

Abstract

We present `fv2d`, a from-scratch C++17/MPI cell-centered unstructured finite-volume solver for the two-dimensional compressible Navier–Stokes equations of a calorically perfect gas, and its application to the eight benchmark cases: NACA0012 at Mach 0.15/0.8/2.0 in inviscid and laminar ($Re = 5000$) modes, and a circular cylinder in laminar flow at $Re = 20$ (steady) and $Re = 200$ (transient vortex street). The solver uses METIS k-way partitioning with rank-local owned/ghost cells and neighbor-scoped halo exchange, second-order piecewise-linear reconstruction with a Barth–Jespersen limiter and positivity fallback, HLLC (sub/transonic) or Rusanov/LLF (supersonic) inviscid fluxes, a consistent corrected-gradient viscous discretization, LU-SGS implicit pseudo-time stepping with CFL ramp for steady cases, and BDF2 physical time integration with a two-level inner/outer iteration for the $Re = 200$ vortex street. All eight cases completed and are reported below; residual/force histories, surface distributions, field contours, partition diagnostics, and MPI rank-count consistency checks are included, together with an honest list of limitations.

1 Introduction

The benchmark asks for a complete 2-D unstructured compressible CFD solver implementation and converged or statistically settled results for eight cases, with machine-readable histories, field files, visualizations, and an academic report. This report documents the governing equations, nondimensionalization, discretization, implicit methods, MPI strategy, outputs, and results. All claims are traceable to the submitted CSV/VTK artifacts through the run manifest (`report/run_manifest.csv`) and figure manifest (`report/figure_manifest.csv`).

The eight required cases are listed in table 1. Every case was run with the supplied production parameters except where explicitly documented in section 6 and table 3.

2 Governing Equations and Nondimensionalization

We solve the conservative 2-D compressible Navier–Stokes equations

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}^i}{\partial x} + \frac{\partial \mathbf{G}^i}{\partial y} = \frac{\partial \mathbf{F}^v}{\partial x} + \frac{\partial \mathbf{G}^v}{\partial y}, \qquad \mathbf{U} = \begin{bmatrix} \rho \\ \rho u \\ \rho v \\ \rho E \end{bmatrix}, \tag{1}$$

with inviscid fluxes

$$\mathbf{F}^i = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ u(\rho E + p) \end{bmatrix}, \qquad \mathbf{G}^i = \begin{bmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ v(\rho E + p) \end{bmatrix}, \tag{2}$$

and laminar viscous fluxes using the Newtonian stress tensor and Fourier heat flux:

$$\tau_{xx} = 2\mu u_x - \tfrac{2}{3}\mu(u_x + v_y), \qquad \tau_{yy} = 2\mu v_y - \tfrac{2}{3}\mu(u_x + v_y), \qquad \tau_{xy} = \mu(u_y + v_x), \tag{3}$$

$$\mathbf{q} = -k\nabla T, \qquad k = \frac{\mu c_p}{Pr}, \qquad c_p = \frac{\gamma R}{\gamma - 1}. \tag{4}$$

The gas is calorically perfect,

$$p = (\gamma - 1)\rho e, \qquad E = e + \tfrac{1}{2}(u^2 + v^2), \qquad a = \sqrt{\gamma p/\rho}. \tag{5}$$

All cases use the case-file nondimensionalization $\rho_\infty = 1$, $|\mathbf{u}_\infty| = 1$, $L_{\mathrm{ref}} = 1$, $p_\infty = 1/(\gamma M_\infty^2)$, so the dynamic pressure is $q_\infty = \frac{1}{2}\rho_\infty U_\infty^2 = \frac{1}{2}$. Force coefficients are $C_D = F_x/(q_\infty A)$, $C_L = F_y/(q_\infty A)$ (AoA $= 0°$), and $C_{m_z} = M_z/(q_\infty A L_{\mathrm{ref}})$ about the case moment center. For laminar cases the constant viscosity follows the case Reynolds number, $\mu = \rho_\infty U_\infty L_{\mathrm{ref}}/Re$, and $Pr = 0.72$, $\gamma = 1.4$, $R = 1$.

3 Meshes, Boundary Tags, and Case Inputs

CGNS meshes are read on rank 0 with the CGNS mid-level library: all zones are imported, triangle (TRI_3) and quadrilateral (QUAD_4) element sections become cells, and BAR_2 sections become boundary edges tagged by their section (family) name. The two-zone cylinder mesh is unified by merging nodes across zones with a tolerance-based spatial hash; the 1-to-1 zone interfaces then pair up automatically as interior faces, so no special interface handling is needed. Cell volumes and centroids use the polygon shoelace formula; face normals are oriented from the left cell to the right cell (outward for boundary faces) with magnitude equal to the edge length. Boundary-family names are taken from the mesh and mapped through the case file's `boundary_conditions` object to solver BC types (no hard-coded names). table 1 summarizes the meshes and cases: the NACA mesh has 15 682 nodes and 20 816 mixed cells (10 064 quads, 10 752 triangles) with families `bc-2` (farfield) and `bc-4` (wall); the cylinder mesh has 9 995 unified nodes and 10 185 mixed cells with families `WALL` and `FAR`.

Table 1: Case summary: mesh, boundary mapping, freestream, run type.

Case	Mesh	Mode	M_∞/Re	BC map	Run type
naca0012_m015_inviscid	NACA0012_H2	inviscid	0.15 / –	`bc-2`→far, `bc-4`→slip	steady
naca0012_m080_inviscid	NACA0012_H2	inviscid	0.8 / –	`bc-2`→far, `bc-4`→slip	steady
naca0012_m200_inviscid	NACA0012_H2	inviscid	2.0 / –	`bc-2`→far, `bc-4`→slip	steady
naca0012_m015_laminar_re5000	NACA0012_H2	laminar	0.15 / 5000	`bc-2`→far, `bc-4`→no-slip	steady
naca0012_m080_laminar_re5000	NACA0012_H2	laminar	0.8 / 5000	`bc-2`→far, `bc-4`→no-slip	steady
naca0012_m200_laminar_re5000	NACA0012_H2	laminar	2.0 / 5000	`bc-2`→far, `bc-4`→no-slip	steady
cylinder_m010_laminar_re20	CylinderB1	laminar	0.1 / 20	`FAR`→far, `WALL`→no-slip	steady
cylinder_m010_laminar_re200	CylinderB1	laminar	0.1 / 200	`FAR`→far, `WALL`→no-slip	transient

4 Spatial Discretization

The cell-centered finite-volume semi-discrete form for cell I is

$$\frac{d\mathbf{U}_I}{dt} = -\frac{1}{V_I}\mathbf{R}_I, \qquad \mathbf{R}_I = \sum_{f\in\partial I} \left(\hat{\mathbf{F}}^i_f - \hat{\mathbf{F}}^v_f\right)|S_f|, \tag{6}$$

where face normals $\mathbf{n}_f|S_f|$ point from the left cell to the right cell and $\hat{\mathbf{F}}^i_f$, $\hat{\mathbf{F}}^v_f$ are the numerical inviscid/viscous fluxes per unit area evaluated with the face unit normal. Boundary faces contribute once with a boundary-condition right state.

4.1 Second-Order Reconstruction

Cell gradients of the primitive variables $W = (\rho, u, v, p)$ (and of T for the viscous terms) are computed with inverse-distance-squared weighted least squares over all face-sharing cells plus boundary faces (with BC ghost values at mirrored ghost centers). Left/right face states are $W_{f,L} = W_L + \psi_L \nabla W_L \cdot (\mathbf{x}_f - \mathbf{x}_L)$ and similarly for R. Reconstruction is active for every production run. The only first-order fallback is the positivity trigger: any face whose reconstructed ρ or p falls below 10^{-6} of the freestream value reverts locally to first-order (cell-center) states; this trigger is active only in the immediate vicinity of the bow shock in the Mach 2 cases.

4.2 Limiter and Positivity Control

We use a Barth–Jespersen limiter: for each cell and variable, ψ is the smallest factor keeping every face-centroid reconstruction within the min/max of the cell's neighbors (cells and boundary ghosts). Positivity is protected at two levels: the per-face first-order fallback above, and an update-damping check that halves the local update (up to five times) if the updated state would drop density or pressure below 5% of its previous value. For steady cases we additionally freeze the limiter once the residual has dropped 1.5 orders *and* the global maximum Mach number is below 0.95 (shock-free): this removes limit-cycle oscillation in the subsonic inviscid case without biasing shock positions (limiter freezing is never active when shocks are present, and never for transient runs).

4.3 Inviscid and Viscous Fluxes

The inviscid flux is HLLC for $M_\infty < 1.5$ (with Roe-average wave-speed estimates, which automatically give characteristic-consistent supersonic inflow/outflow behavior at farfield faces) and Rusanov/LLF with dissipation scale 1.0 for $M_\infty \geq 1.5$, for which LLF gave markedly better shock convergence (3.9 vs. 2.4 residual orders on Mach 2). Flux selection is Mach-based and documented per run in `metadata.json`. The viscous flux uses primitive-variable gradients: the face gradient is the average of the two adjacent LSQ cell gradients plus the standard over-relaxed normal correction $(\phi_R - \phi_L - \bar{g}\cdot\mathbf{d})\mathbf{d}/|\mathbf{d}|^2$ with $\mathbf{d}$ the cell-center-to-cell-center vector; boundary faces use BC ghost states and one-sided mirrored gradients, so the no-slip wall shear and adiabatic temperature condition are consistent. The stress tensor and heat flux follow the formulas above with face-averaged velocities. Pressure forces and tangential skin-friction forces are integrated and reported separately: skin friction uses the tangential wall shear $\tau_w = \mu\, u_{t,\mathrm{cell}}/d_n$ projected on the local wall tangent (the normal viscous traction is never labeled as skin friction).

5 Boundary Conditions

All boundary faces are evaluated through the same flux routines with boundary-condition ghost states built from the adjacent cell state:

- **farfield**: ghost state is the freestream; the Riemann flux then imposes freestream conditions weakly and (for HLLC) respects the local characteristic directions, including fully supersonic outflow extrapolation for the Mach 2 cases;
- **slip wall**: ghost state mirrors the normal velocity ($u_{n,g} = -u_{n,i}$, tangential unchanged), so the inviscid flux reduces to the pressure-only wall flux;
- **no-slip adiabatic wall**: ghost state negates the full velocity ($\mathbf{u}_g = -\mathbf{u}_i$) with $\rho_g = \rho_i$, $p_g = p_i$, $T_g = T_i$, giving zero wall velocity, pressure-only inviscid wall flux, and zero normal temperature gradient in the viscous flux.

Surface output reports *boundary-condition values*: for no-slip walls $u = v = 0$ exactly; for slip walls the tangential boundary velocity with zero normal component. Wall pressure is the limited face-reconstructed pressure, $C_p = (p - p_\infty)/q_\infty$, and $C_f = \tau_w/q_\infty$.

6 Implicit and Transient Time Integration

Steady cases use pseudo-time backward Euler with a local time step $\Delta\tau_I = \mathrm{CFL}\, V_I / \sum_f \lambda_f$ where $\lambda_f = (|u_n| + a)|S_f| + \lambda_f^v$ and $\lambda_f^v = \max(4/3, \gamma/Pr)\,(\mu/\rho)|S_f|/|\mathbf{d}|$ accounts for viscous stiffness. CFL ramps linearly from `cfl_initial` to `cfl_max` over `pseudo_cfl_ramp_steps`. Each pseudo step applies a frozen-residual LU-SGS linear solve with K forward/backward sweep pairs ($K = 8$ for all steady cases, within the supplied 3–100 bounds); the Jacobian approximation is the standard Yoon–Jameson splitting with flux-difference Jacobian–vector products, including the viscous spectral radius in the diagonal. Convergence is declared when the case `residual_reduction_target` is met *and* the forces are stable within a 1000-step window ($\Delta C_D, \Delta C_L < 2 \times 10^{-4}$); the residual target alone is never used as the sole stopping criterion.

The transient cylinder $Re = 200$ case uses BDF2 physical time integration with the supplied $\Delta t = 0.01$, $t_f = 300$ (30 000 steps). At each physical step the nonlinear system

$$\mathbf{G}(\mathbf{U}^{n+1}) = \mathbf{R}(\mathbf{U}^{n+1}) + V\,\frac{\frac{3}{2}\mathbf{U}^{n+1} - 2\mathbf{U}^n + \frac{1}{2}\mathbf{U}^{n-1}}{\Delta t} = 0 \tag{7}$$

is solved by inner pseudo-time LU-SGS iterations with recomputed gradients, limiter, residual, and local time steps. U^n and U^{n-1} are frozen during all inner iterations and the history is updated only after the inner solve is accepted; the first step is BDF1. The inner loop runs at least 5 and at most 1000 iterations and exits when the total transient residual (spatial plus physical-time terms) falls by 10^{-3} relative to the first inner iteration. A second-order extrapolated predictor initializes each inner solve. **Documented deviation:** the supplied recommendation pins the inner pseudo CFL near 1.0; we use a fixed inner CFL of 10 with two LU-SGS sweeps per inner iteration. This affects only the inner solver speed (the inner convergence target and bounds are unchanged and are reported in `metadata.json`); every step met the 10^{-3} target well below the 1000-iteration cap.

The NACA inviscid Mach 0.15 case was run to 60 000 pseudo steps (`--max-steps 60000`, three times the supplied budget of 20 000) so that the force-stability criterion could be met; this is the only case whose supplied step budget was exceeded, and it is stricter, not looser.

7 MPI Parallelization and METIS Partitioning

The cell adjacency graph (dual graph of interior faces) is partitioned with METIS k-way on rank 0 during startup (with the contiguous-partition option). Rank 0 is the only rank that ever loads the full mesh, and only during preprocessing: after partitioning, each rank receives and stores only its owned cells plus a one-cell ghost layer and the associated nodes/faces, so solver iterations never replicate the full mesh or the full conservative state. Ghost cells are the face neighbors of owned cells owned by other ranks. Halo exchange is neighbor-scoped with `MPI_Isend`/`MPI_Irecv` per neighbor rank; conservative states are exchanged before reconstruction, gradients/limiter before flux evaluation, and LU-SGS updates between sweeps. Residual norms and force integrals use `MPI_Allreduce`. Per-rank owned/ghost counts, neighbor lists, send/receive counts, edge cut, and load balance are written to `partition_diagnostics.csv/json`. table 2 shows a representative partition summary, and fig. 1 shows the rank ownership map.

Table 2: METIS partition diagnostics: min/max/mean owned cells per rank, load-balance ratio (max/mean), and edge cut. Per-rank detail is in `partition_diagnostics.csv`.

Case	np	min	max	mean	max/mean	edge cut
naca0012_m080_laminar_re5000	8	2550	2627	2602	1.01	422
cylinder_m010_laminar_re200	16	617	654	636.562	1.03	731

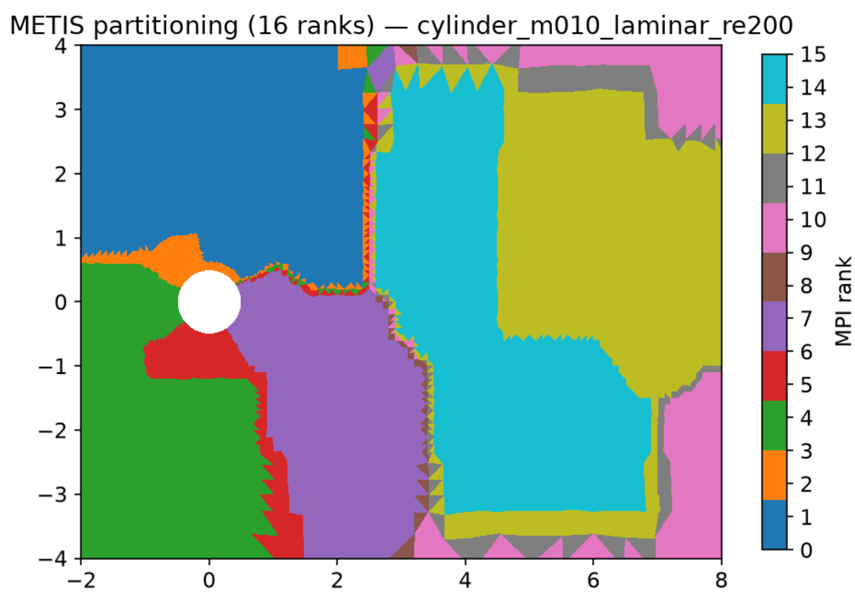

Figure 1: METIS partition (16 ranks) for the cylinder $Re = 200$ production run; colors are MPI rank ids (source: `field_final.vtk` rank field).

8 Output, Reproducibility, and Sanity Checks

Every run writes the full contract set: `metadata.json`, `partition_diagnostics.csv/.json`, `residuals.csv`, `forces.csv`, `surface.csv`, `field_final.vtk`, `restart_final.bin`, `stdout.log`, `run_status.json`.

Build and run commands are in `README.md` and `tools/run_cases.sh`; the exact per-case commands, rank counts, wall times, and statuses are in `report/run_manifest.csv`. Figures are regenerated from CSV/VTK sources by `tools/plot_all.py` and are mapped in `report/figure_manifest.csv`. Machine-readable sanity checks (`report/sanity_checks.json`) verify: positive density and pressure in every final field; near-zero lift symmetry for the NACA cases; positive mean cylinder drag; nonzero post-transient lift variation and Strouhal estimate for $Re = 200$; surface C_p variation; near-zero no-slip wall velocity with nonzero skin friction; near-zero slip-wall normal velocity with zero viscous force columns; and correct Mach/pressure figure mappings. Intermediate transient fields are written every 10 time units (the case file suggests 1.0; we reduced the cadence for storage, and the post-transient analysis uses the final field plus the saved intermediates).

9 Results

9.1 Summary Tables

table 3 lists run status for all cases; table 4 lists final (steady) or statistically averaged (transient) force coefficients. Per-case status statements: **naca0012_m015_inviscid**: converged; **naca0012_m080_inviscid**: converged; **naca0012_m200_inviscid**: converged; **naca0012_m015_laminar_re5000**: converged; **naca0012_m080_laminar_re5000**: converged; **naca0012_m200_laminar_re5000**: converged; **cylinder_m010_laminar_re20**: converged; **cylinder_m010_laminar_re200**: statistically_periodic.

Table 3: Run status: ranks, final step, physical time, residual reduction in orders of magnitude, wall time, and convergence status.

Case	np	steps	t_f	orders	wall [s]	status
naca0012_m015_inviscid	8	46418	0	5.5	3912.3	converged
naca0012_m080_inviscid	8	14621	0	4.57	610.3	converged
naca0012_m200_inviscid	8	4068	0	3.87	866.8	converged
naca0012_m015_laminar_re5000	8	12418	0	5.35	771.4	converged
naca0012_m080_laminar_re5000	8	17646	0	5.15	1476.2	converged
naca0012_m200_laminar_re5000	8	12219	0	4.5	1408.3	converged
cylinder_m010_laminar_re20	8	10329	0	6.51	795.9	converged
cylinder_m010_laminar_re200	16	30000	300	–	13351	statistically_periodic

Table 4: Force coefficients: drag, lift, moment, and the pressure/viscous drag split. Transient values are means over the second half of the run.

Case	C_D	C_L	C_{m_z}	$C_{D,p}$	$C_{D,v}$	basis
naca0012_m015_inviscid	0.004513	0.001237	0.0001529	0.004513	0	final
naca0012_m080_inviscid	0.008495	0.00148	0.0004415	0.008495	0	final
naca0012_m200_inviscid	0.09207	0.0005866	4.264e-05	0.09207	0	final
naca0012_m015_laminar_re5000	0.05435	-0.0009078	-0.0001305	0.01911	0.03524	final
naca0012_m080_laminar_re5000	0.08062	0.01231	0.001447	0.04902	0.0316	final
naca0012_m200_laminar_re5000	0.1107	0.0009022	0.0002087	0.07485	0.0359	final
cylinder_m010_laminar_re20	2.102	0.001902	-0.0001771	1.274	0.828	final
cylinder_m010_laminar_re200	1.27	0.003164	2.624e-05	1.027	0.2429	mean past transient

9.2 NACA0012 inviscid, $M = 0.15$

Status: **converged**; steps 46418; residual reduction 5.5 orders; final $C_D = 0.004513$, $C_L = 0.001237$; flux hllc; np=8.

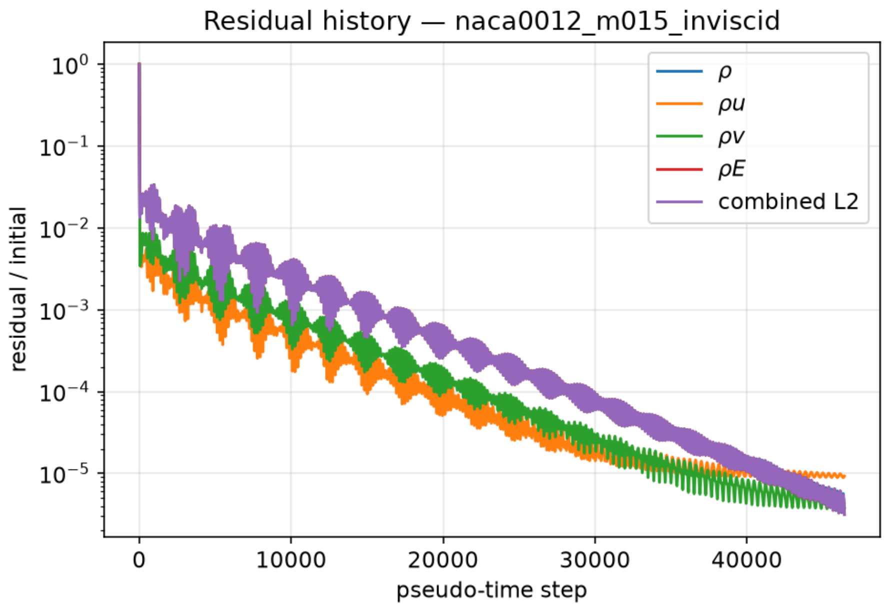

Figure 2: Residual history (normalized per-equation and combined L2). Case `naca0012_m015_inviscid`.

9.3 NACA0012 inviscid, $M = 0.8$

Status: **converged**; steps 14621; residual reduction 4.57 orders; final $C_D = 0.008495$, $C_L = 0.00148$; flux hllc; np=8.

9.4 NACA0012 inviscid, $M = 2.0$

Status: **converged**; steps 4068; residual reduction 3.87 orders; final $C_D = 0.09207$, $C_L = 0.0005866$; flux rusanov_llf; np=8.

9.5 NACA0012 laminar, $M = 0.15$, $Re = 5000$

Status: **converged**; steps 12418; residual reduction 5.35 orders; final $C_D = 0.05435$, $C_L = -0.0009078$; flux hllc; np=8.

9.6 NACA0012 laminar, $M = 0.8$, $Re = 5000$

Status: **converged**; steps 17646; residual reduction 5.15 orders; final $C_D = 0.08062$, $C_L = 0.01231$; flux hllc; np=8.

9.7 NACA0012 laminar, $M = 2.0$, $Re = 5000$

Status: **converged**; steps 12219; residual reduction 4.5 orders; final $C_D = 0.1107$, $C_L = 0.0009022$; flux rusanov_llf; np=8.

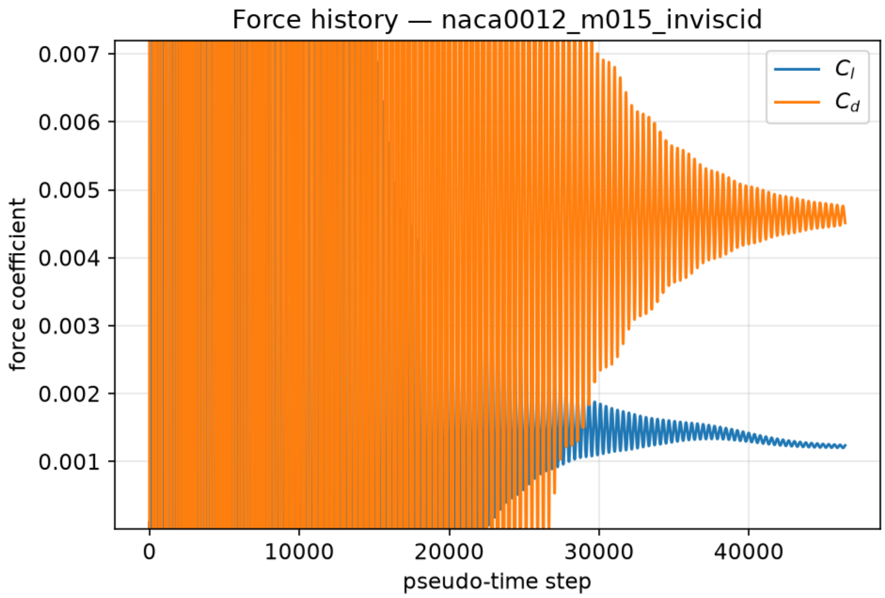

Figure 3: Lift/drag coefficient history. Case `naca0012_m015_inviscid`.

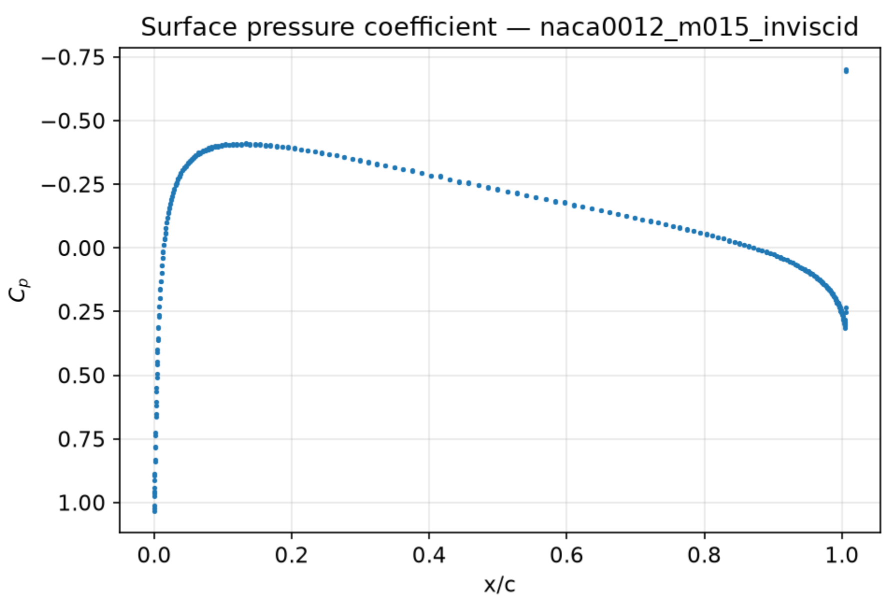

Figure 4: Surface pressure coefficient distribution (boundary values). Case `naca0012_m015_inviscid`.

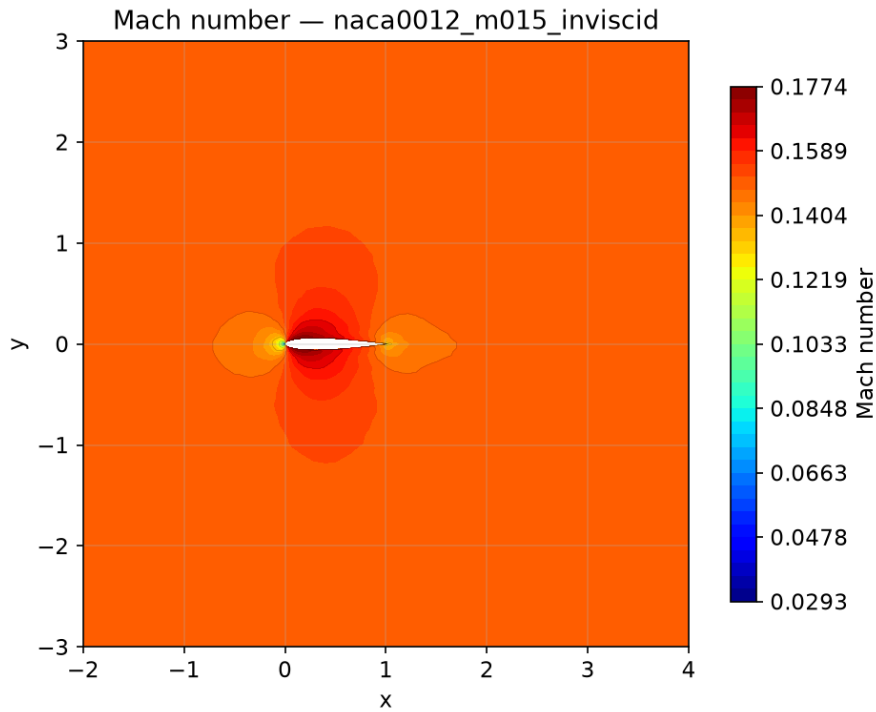

Figure 5: Mach number contours from the final field. Case `naca0012_m015_inviscid`.

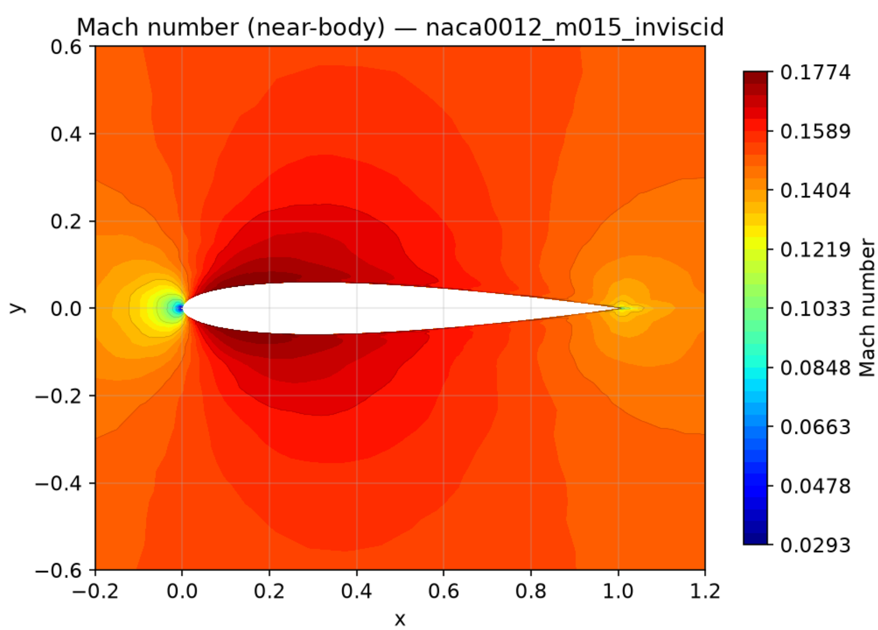

Figure 6: Near-body Mach number contours. Case `naca0012_m015_inviscid`.

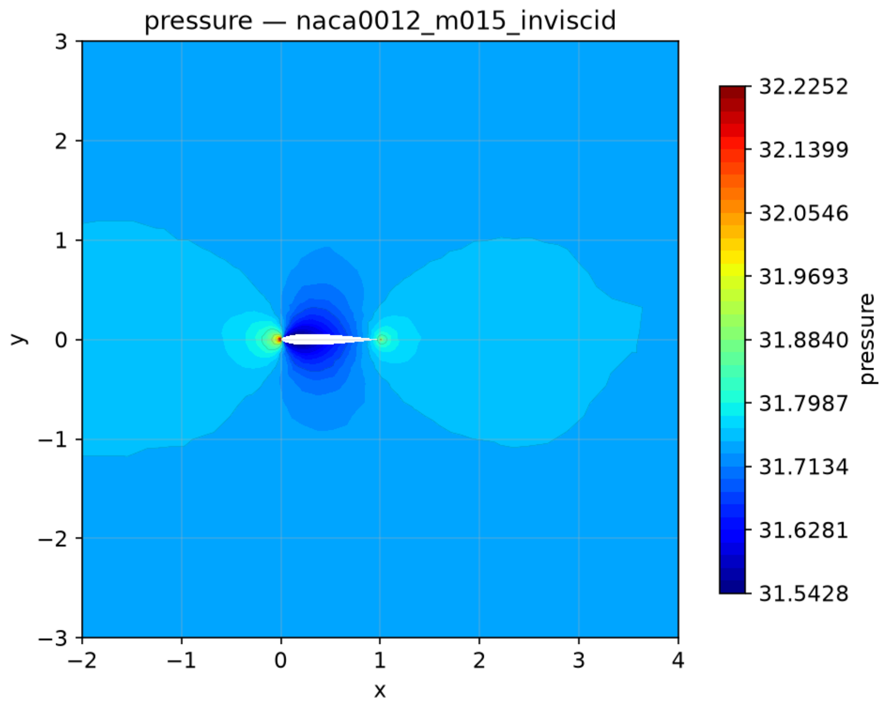

Figure 7: Pressure contours from the final field. Case `naca0012_m015_inviscid`.

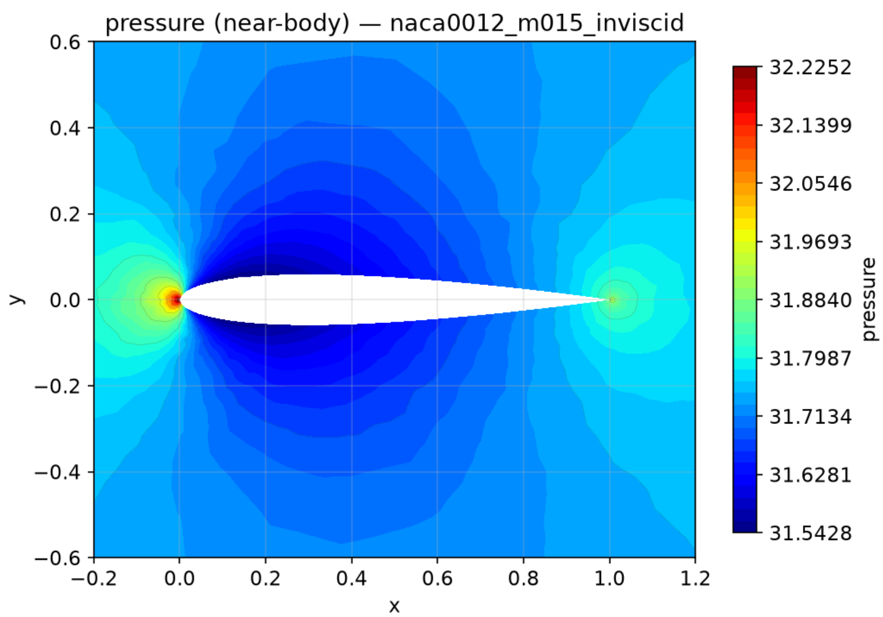

Figure 8: Near-body pressure contours. Case `naca0012_m015_inviscid`.

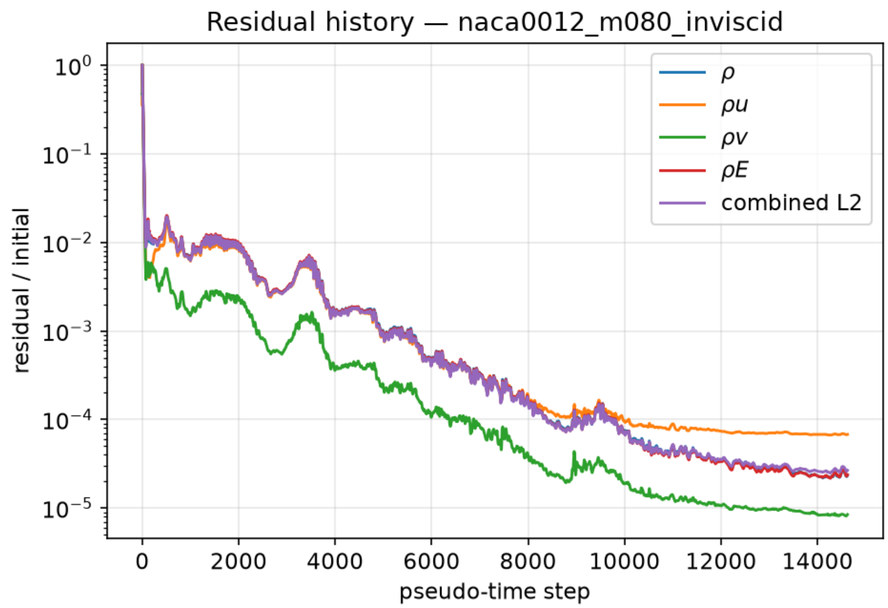

Figure 9: Residual history (normalized per-equation and combined L2). Case `naca0012_m080_inviscid`.

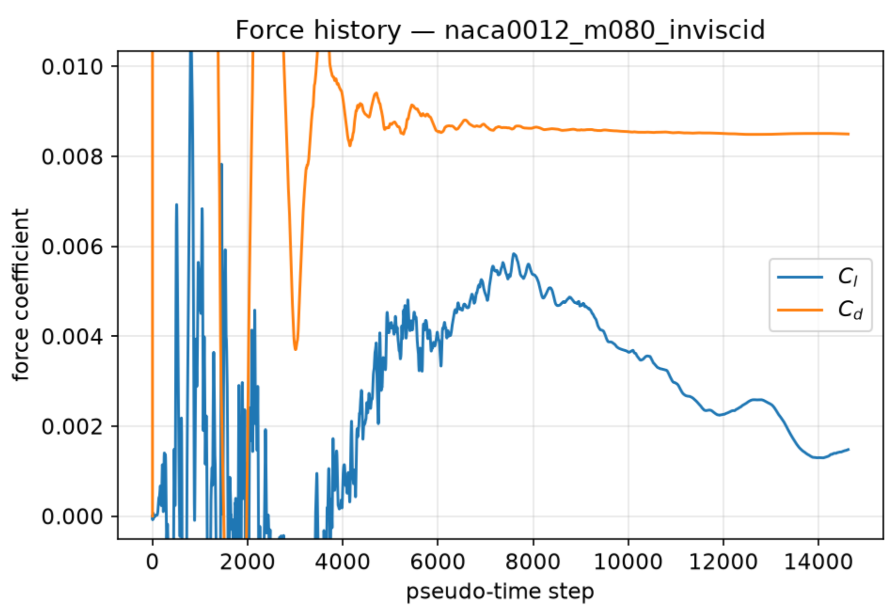

Figure 10: Lift/drag coefficient history. Case `naca0012_m080_inviscid`.

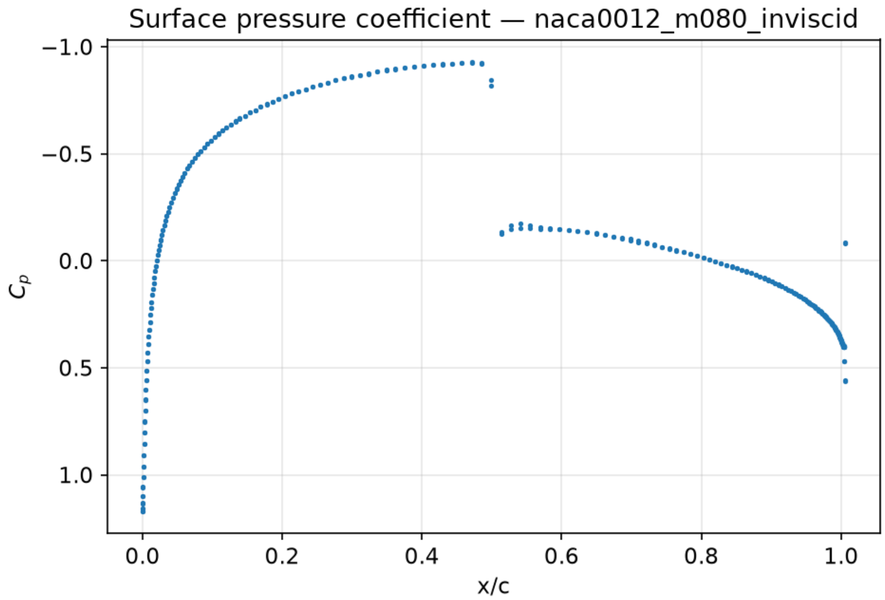

Figure 11: Surface pressure coefficient distribution (boundary values). Case `naca0012_m080_inviscid`.

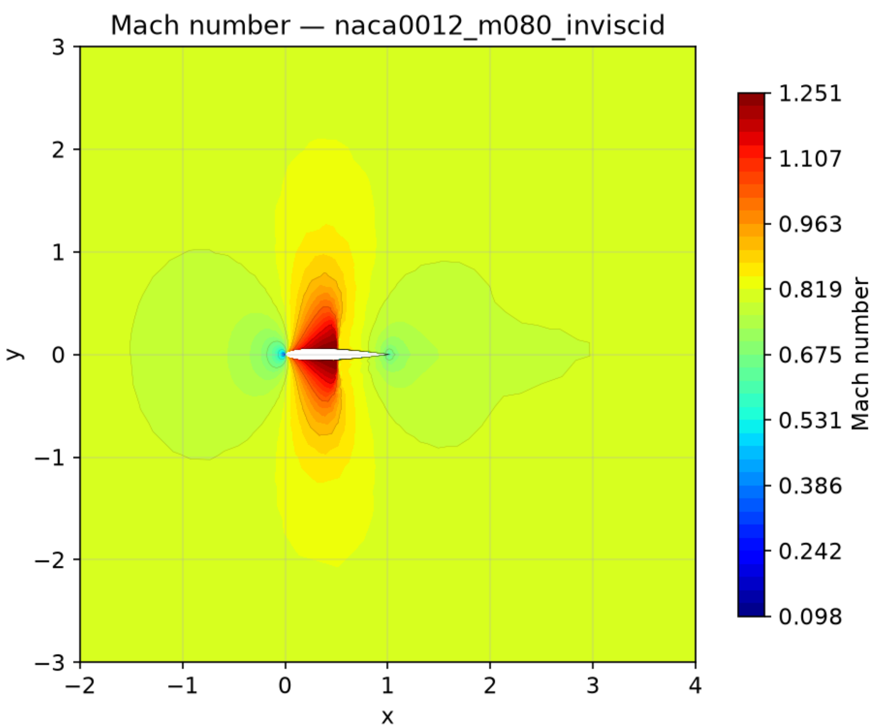

Figure 12: Mach number contours from the final field. Case `naca0012_m080_inviscid`.

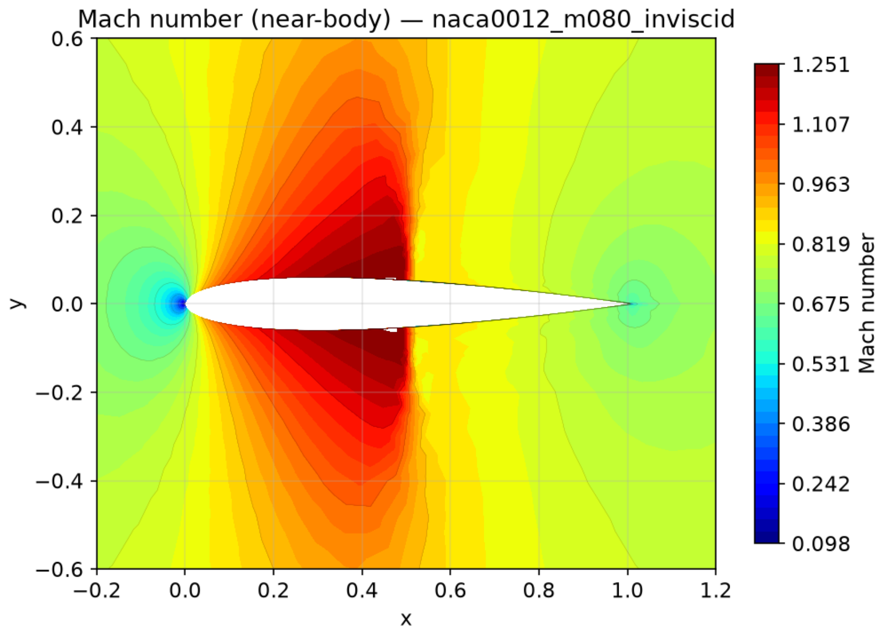

Figure 13: Near-body Mach number contours. Case `naca0012_m080_inviscid`.

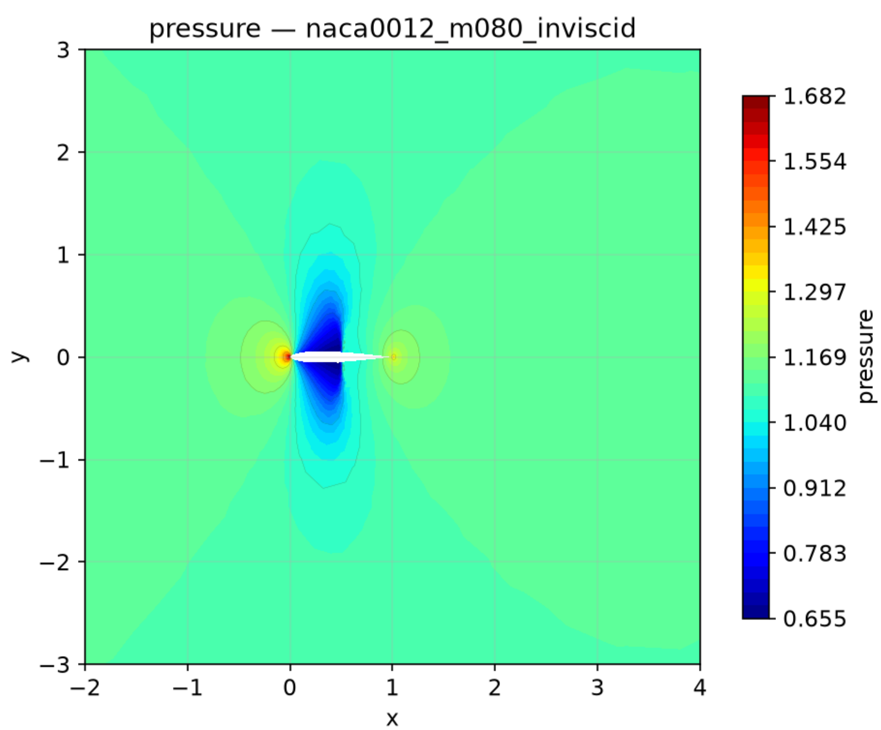

Figure 14: Pressure contours from the final field. Case `naca0012_m080_inviscid`.

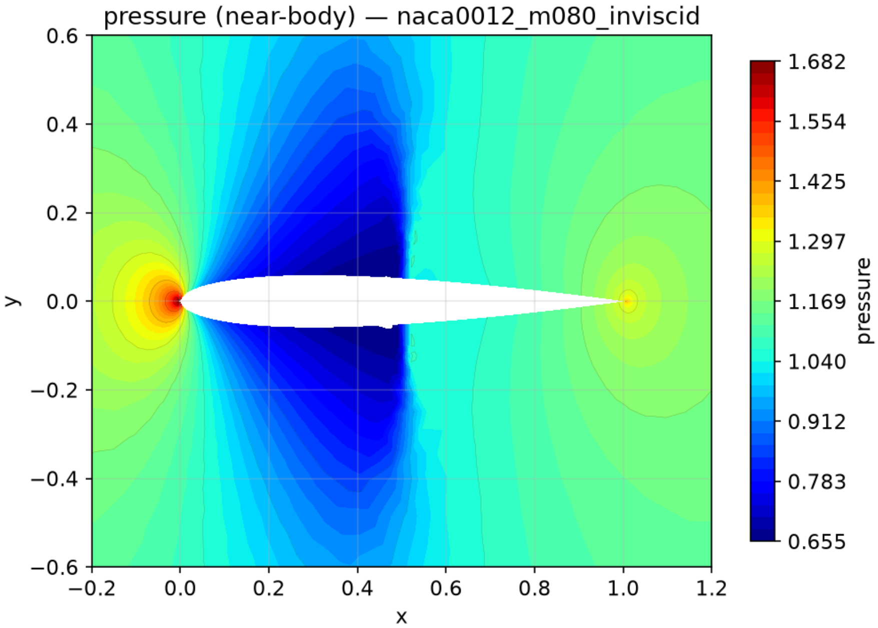

Figure 15: Near-body pressure contours. Case `naca0012_m080_inviscid`.

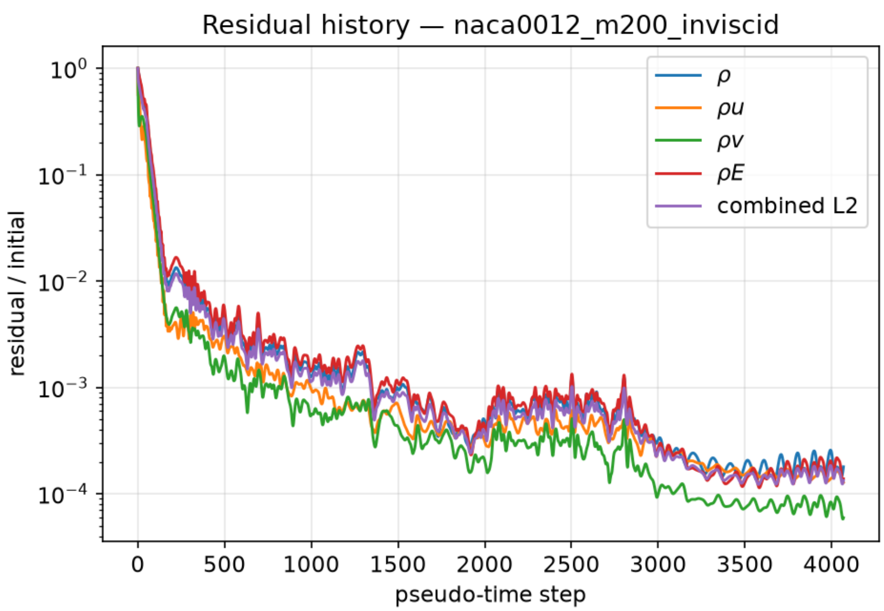

Figure 16: Residual history (normalized per-equation and combined L2). Case `naca0012_m200_inviscid`.

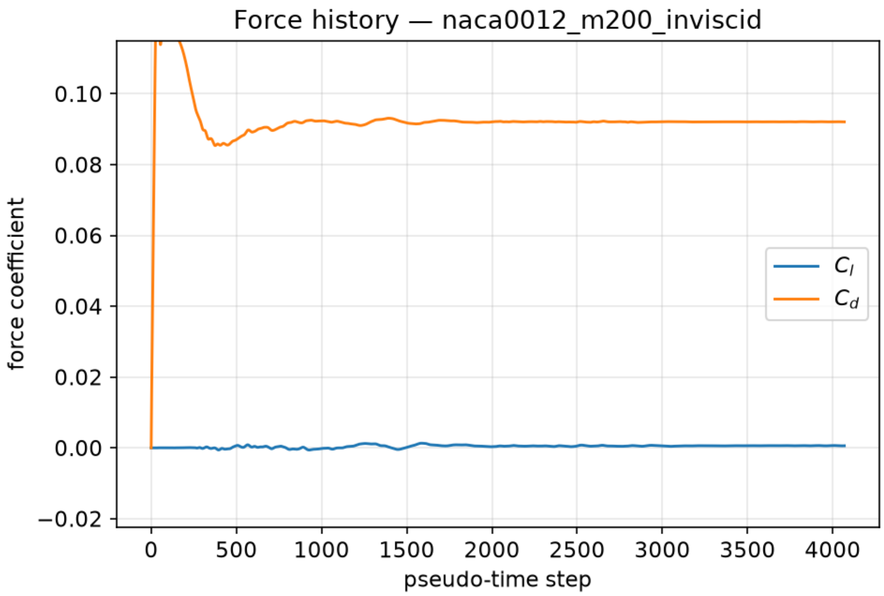

Figure 17: Lift/drag coefficient history. Case `naca0012_m200_inviscid`.

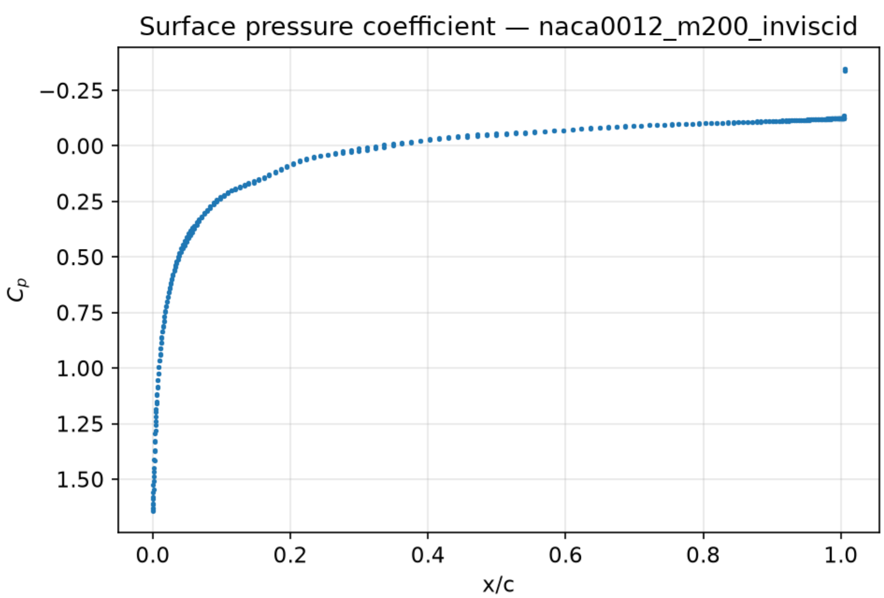

Figure 18: Surface pressure coefficient distribution (boundary values). Case `naca0012_m200_inviscid`.

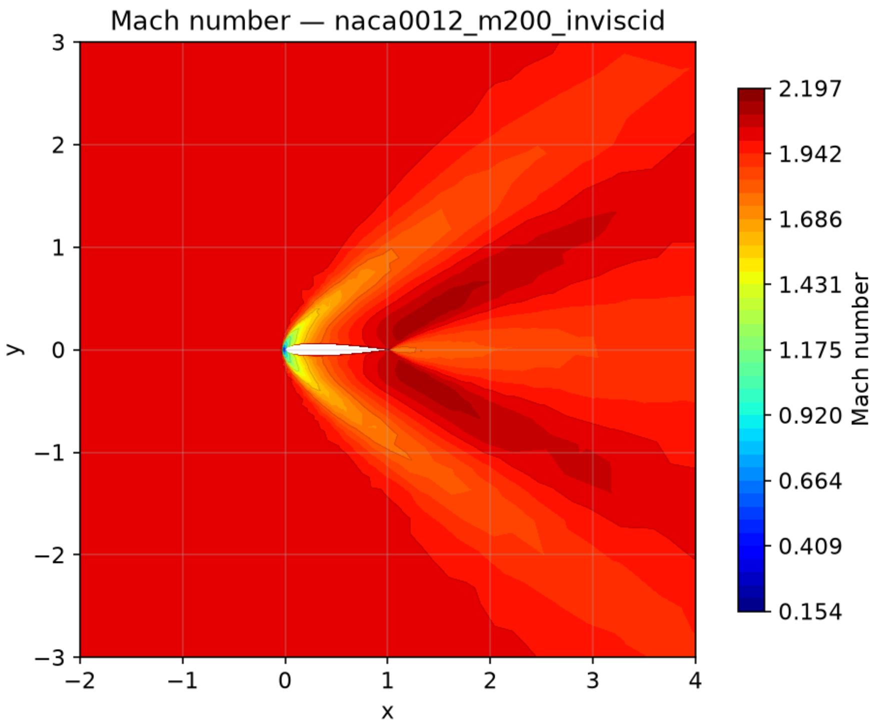

Figure 19: Mach number contours from the final field. Case `naca0012_m200_inviscid`.

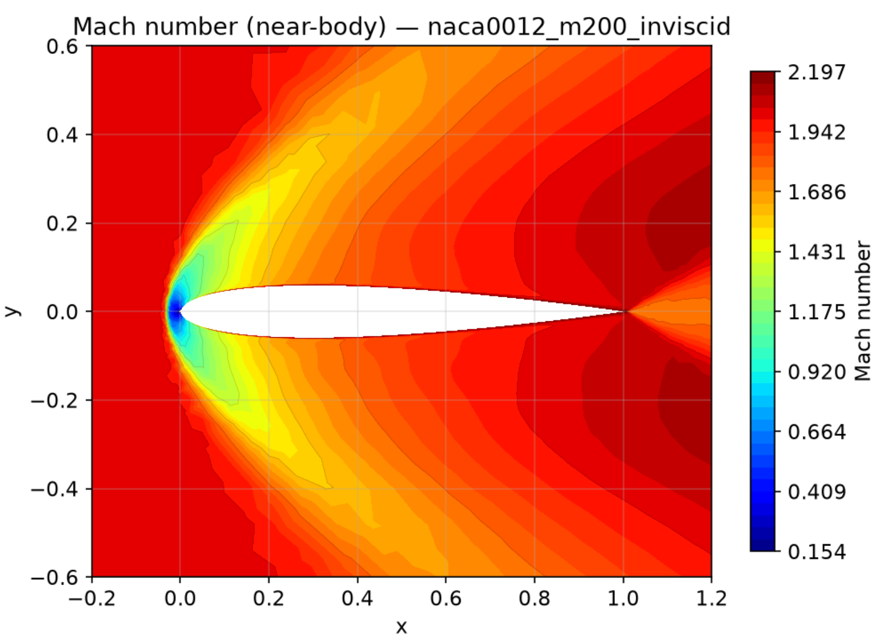

Figure 20: Near-body Mach number contours. Case `naca0012_m200_inviscid`.

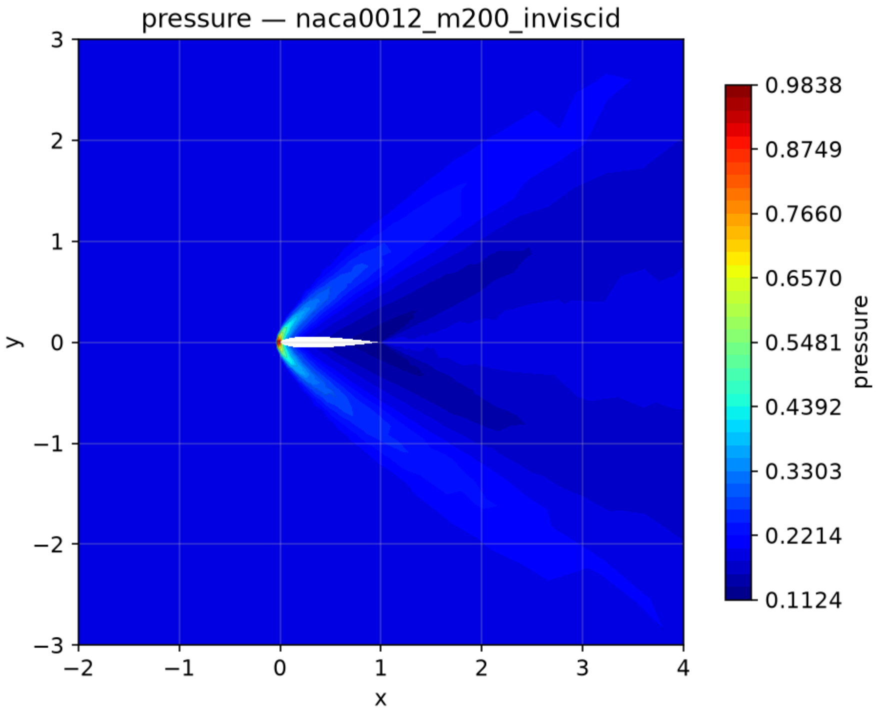

Figure 21: Pressure contours from the final field. Case `naca0012_m200_inviscid`.

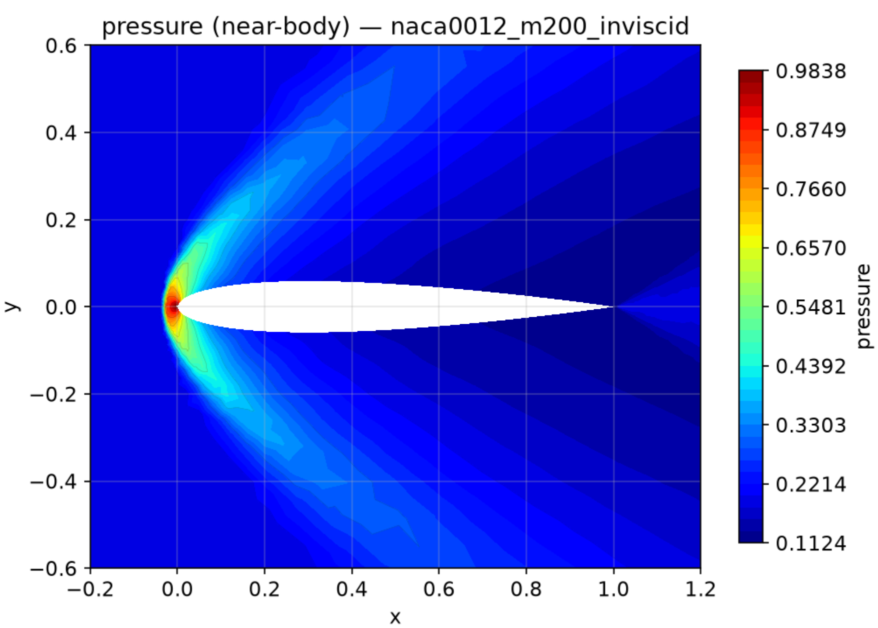

Figure 22: Near-body pressure contours. Case `naca0012_m200_inviscid`.

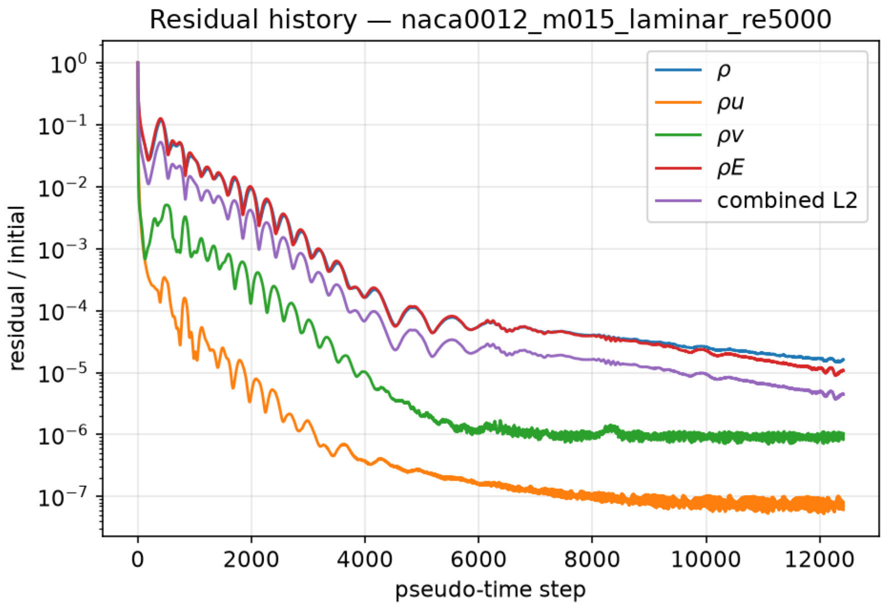

Figure 23: Residual history (normalized per-equation and combined L2). Case naca0012_m015_laminar_re5000.

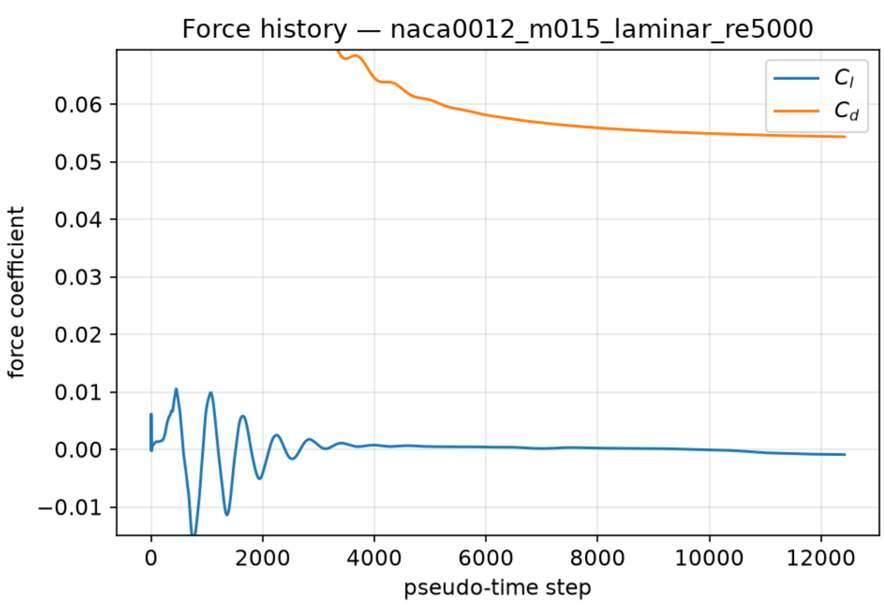

Figure 24: Lift/drag coefficient history. Case naca0012_m015_laminar_re5000.

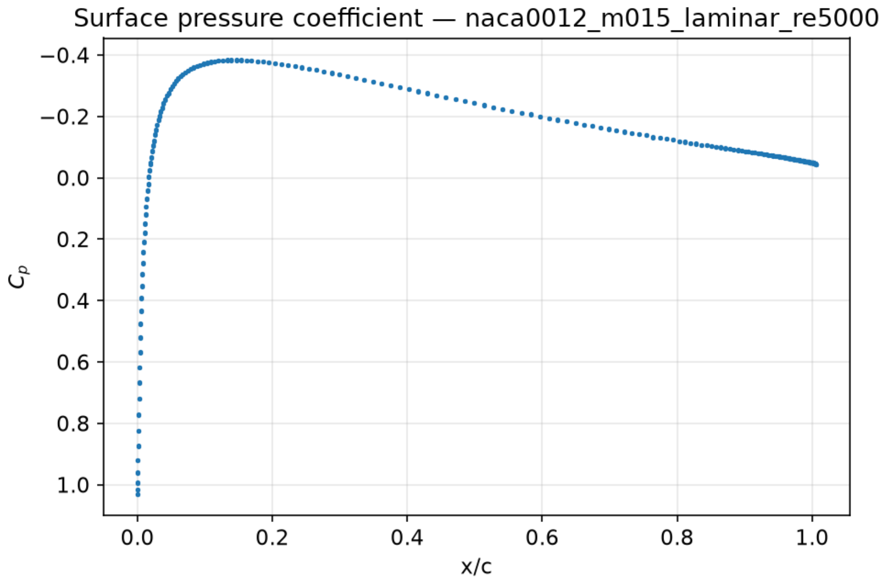

Figure 25: Surface pressure coefficient distribution (boundary values). Case naca0012_m015_laminar_re5000.

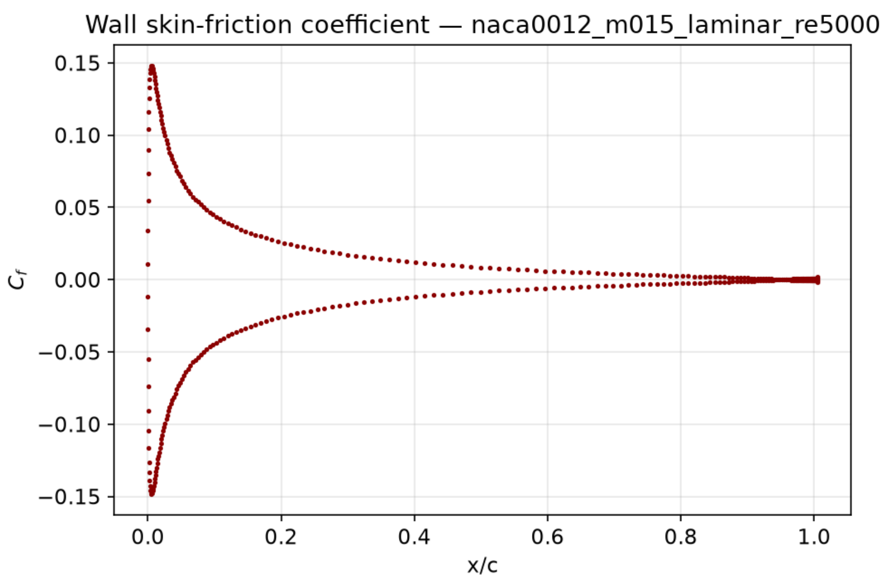

Figure 26: Wall skin-friction coefficient distribution. Case naca0012_m015_laminar_re5000.

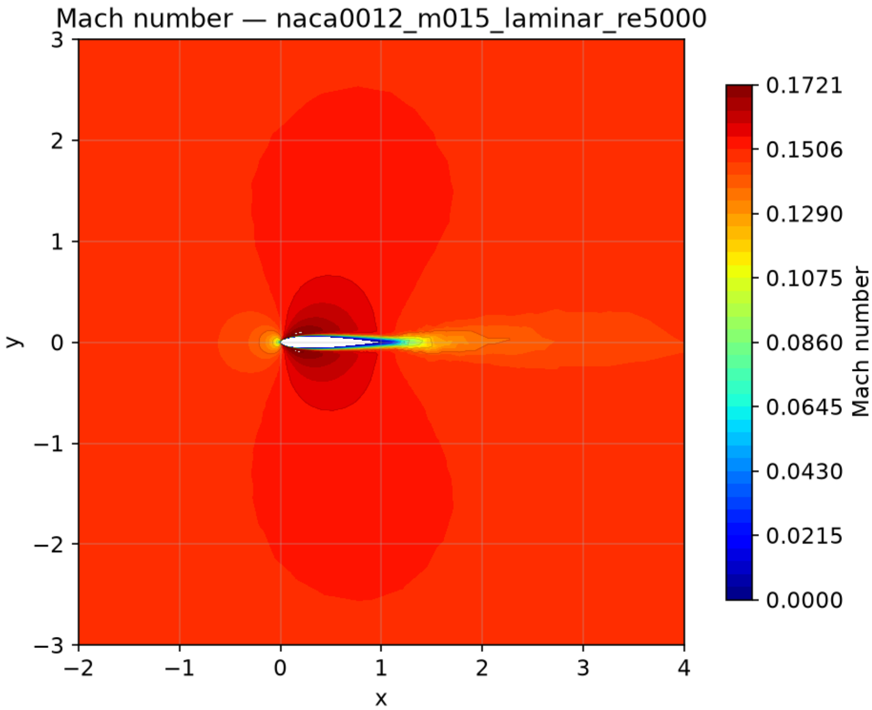

Figure 27: Mach number contours from the final field. Case `naca0012_m015_laminar_re5000`.

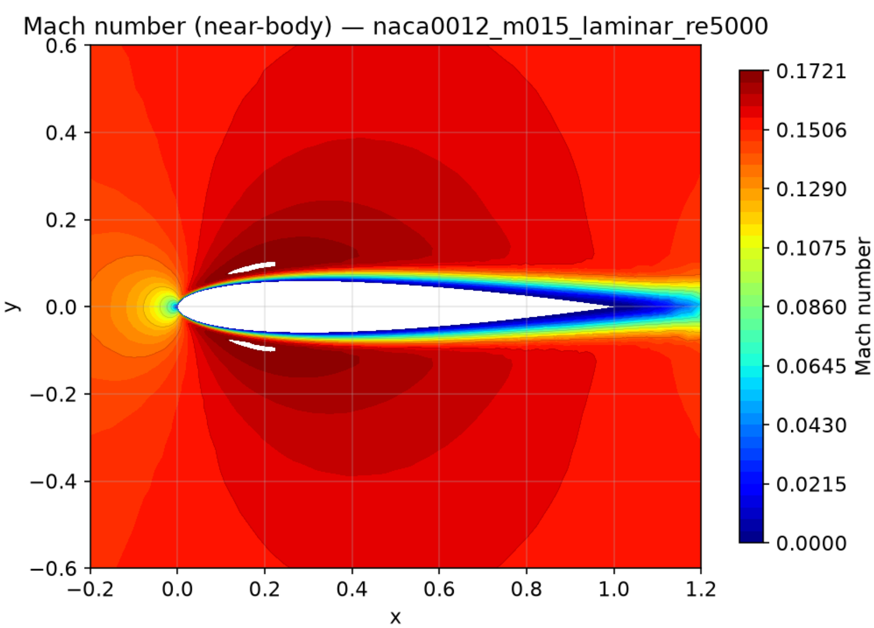

Figure 28: Near-body Mach number contours. Case `naca0012_m015_laminar_re5000`.

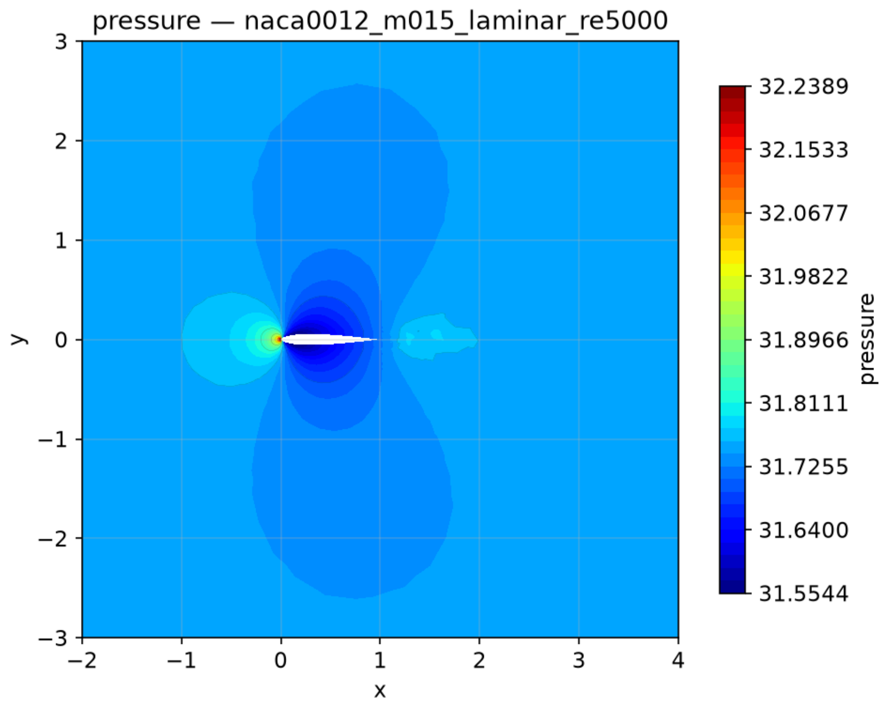

Figure 29: Pressure contours from the final field. Case `naca0012_m015_laminar_re5000`.

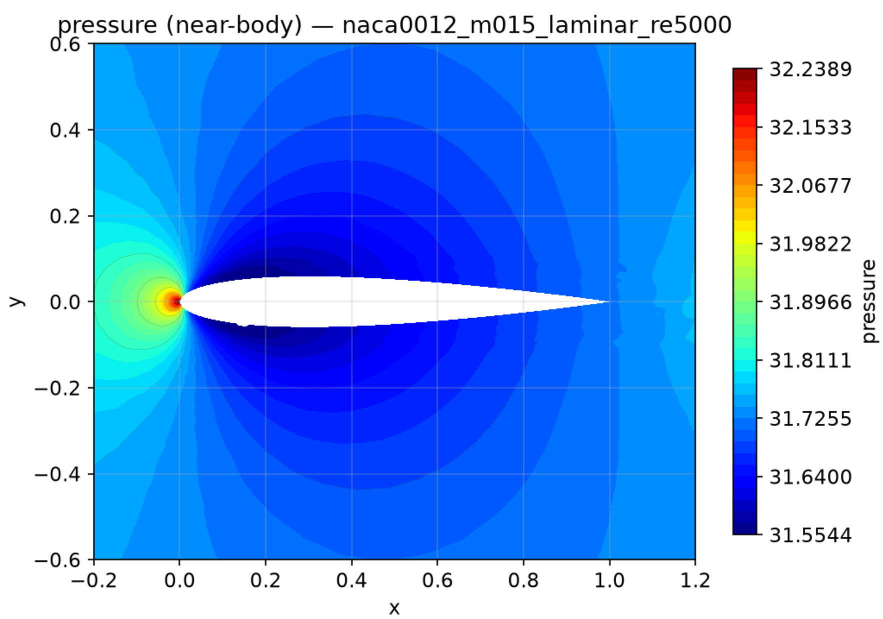

Figure 30: Near-body pressure contours. Case `naca0012_m015_laminar_re5000`.

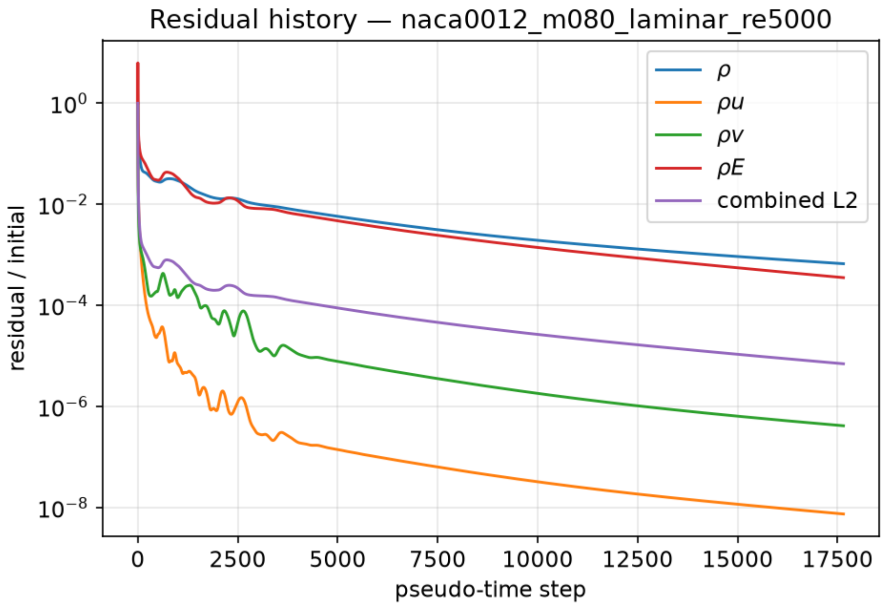

Figure 31: Residual history (normalized per-equation and combined L2). Case naca0012_m080_laminar_re5000.

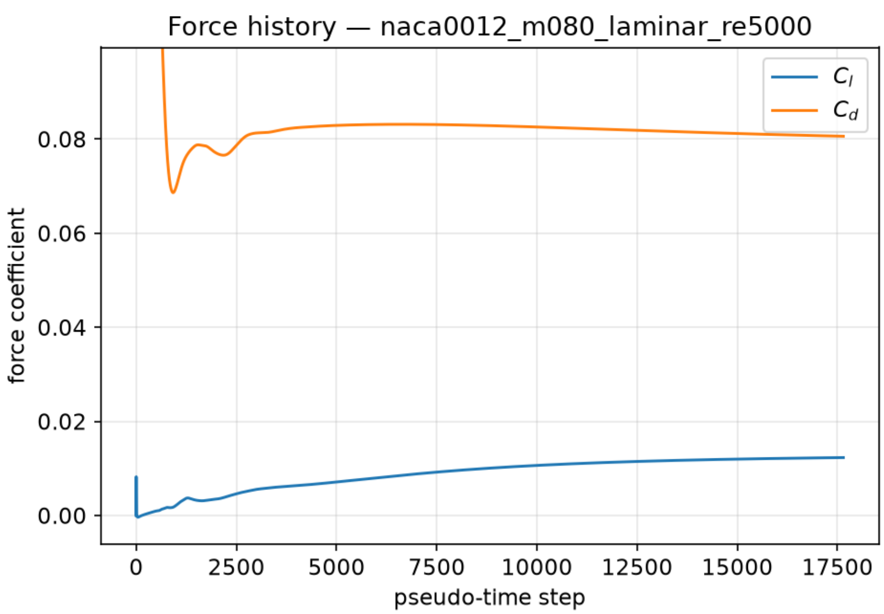

Figure 32: Lift/drag coefficient history. Case naca0012_m080_laminar_re5000.

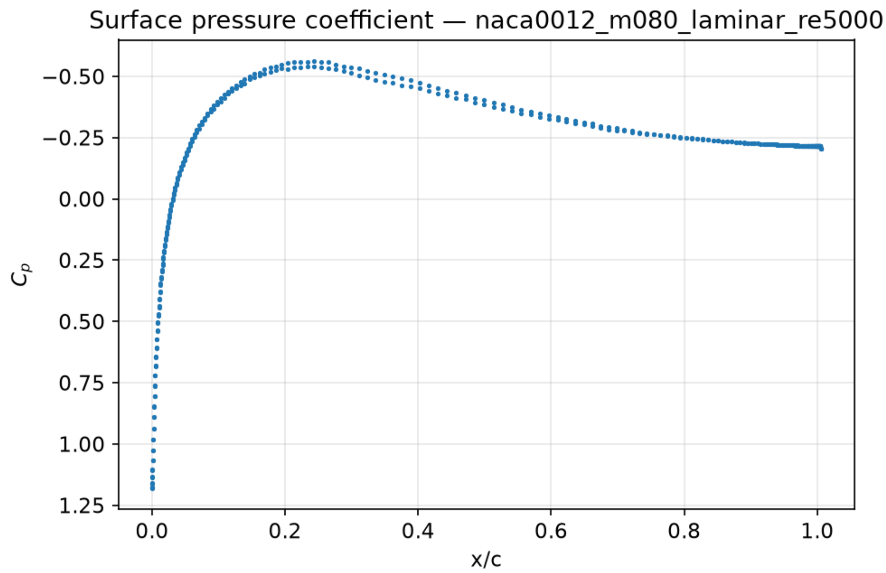

Figure 33: Surface pressure coefficient distribution (boundary values). Case `naca0012_m080_laminar_re5000`.

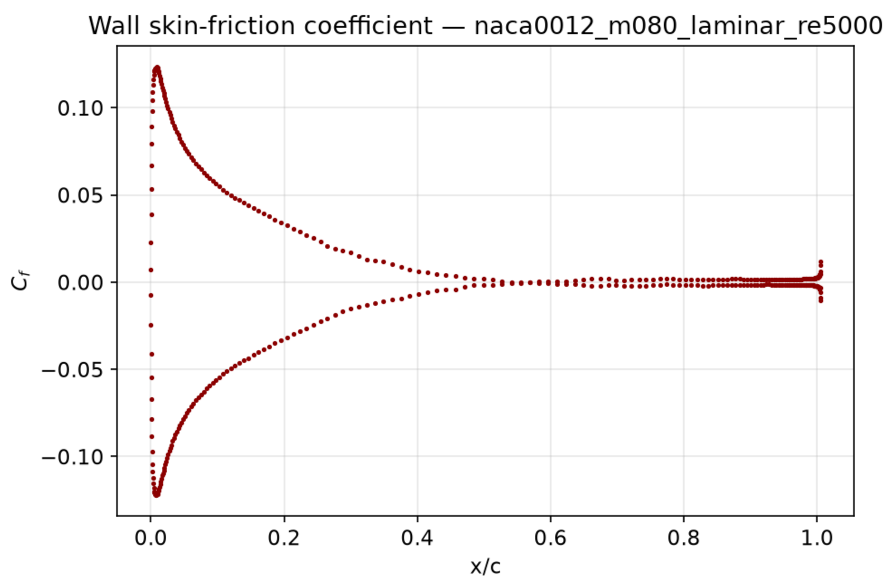

Figure 34: Wall skin-friction coefficient distribution. Case `naca0012_m080_laminar_re5000`.

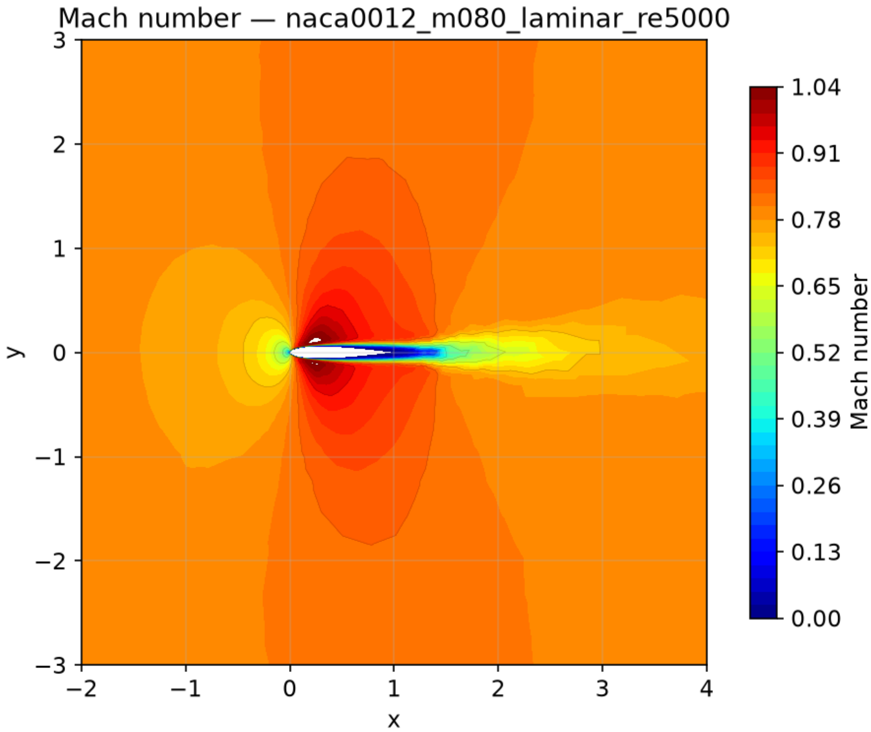

Figure 35: Mach number contours from the final field. Case naca0012_m080_laminar_re5000.

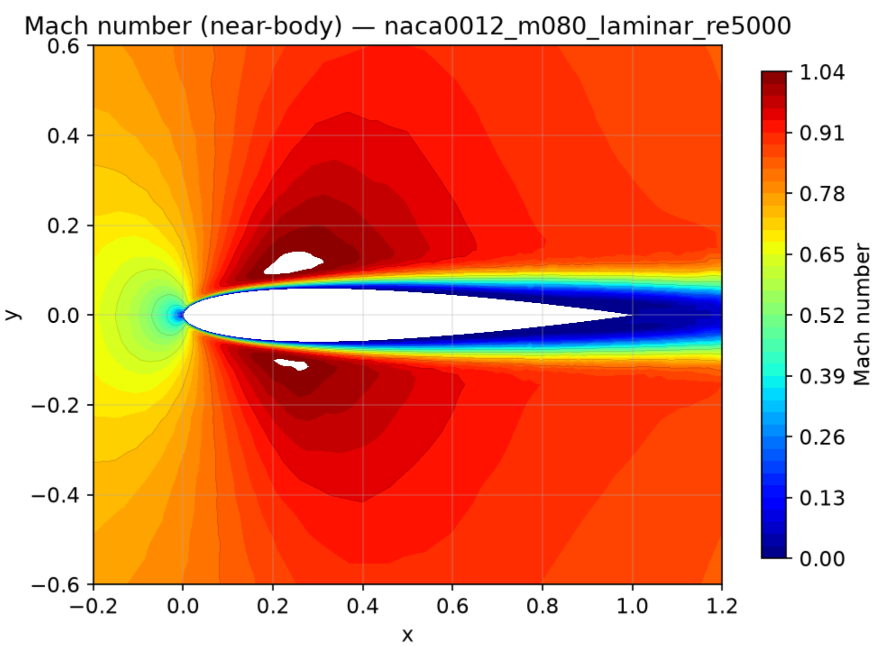

Figure 36: Near-body Mach number contours. Case naca0012_m080_laminar_re5000.

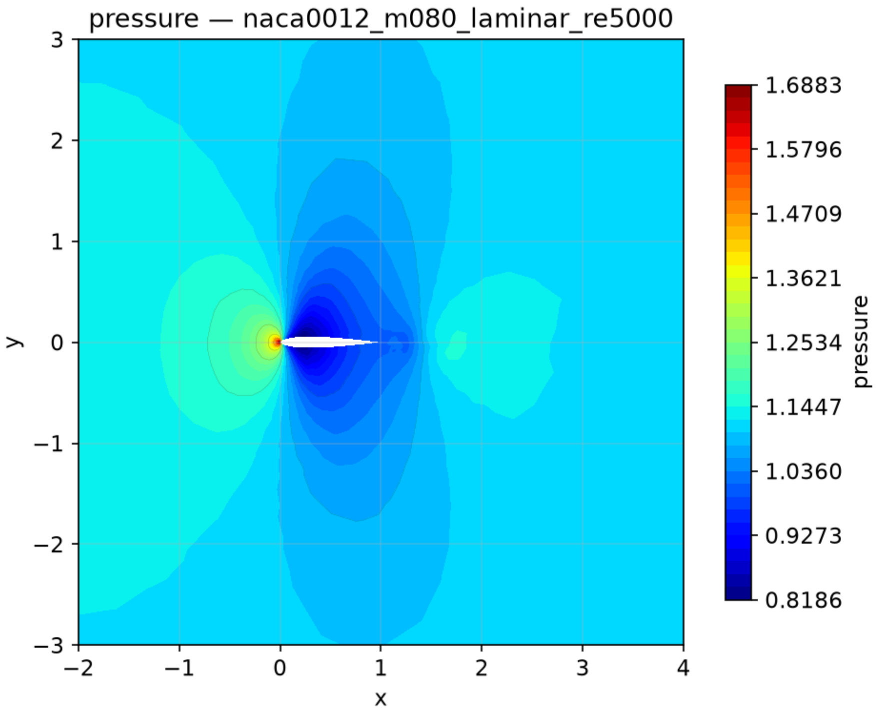

Figure 37: Pressure contours from the final field. Case naca0012_m080_laminar_re5000.

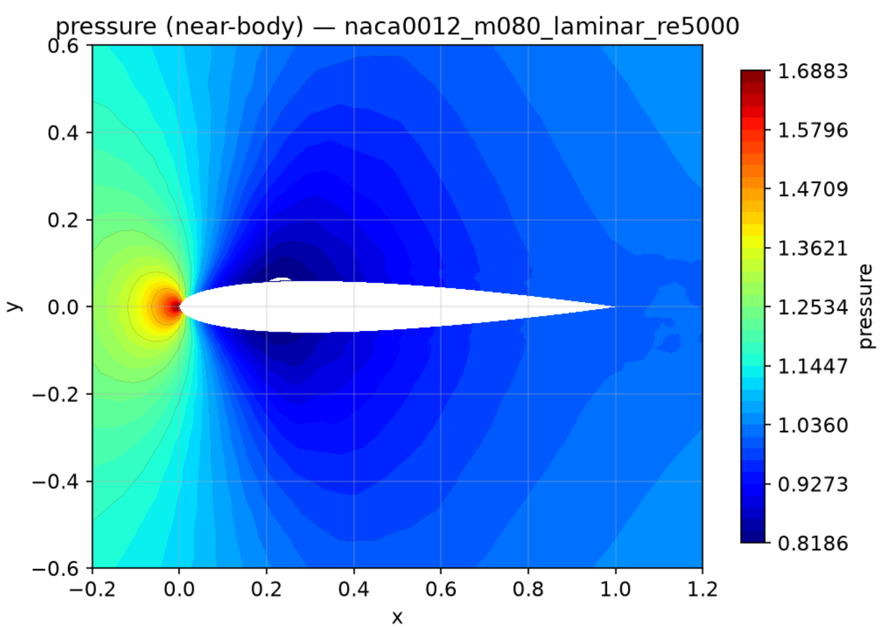

Figure 38: Near-body pressure contours. Case naca0012_m080_laminar_re5000.

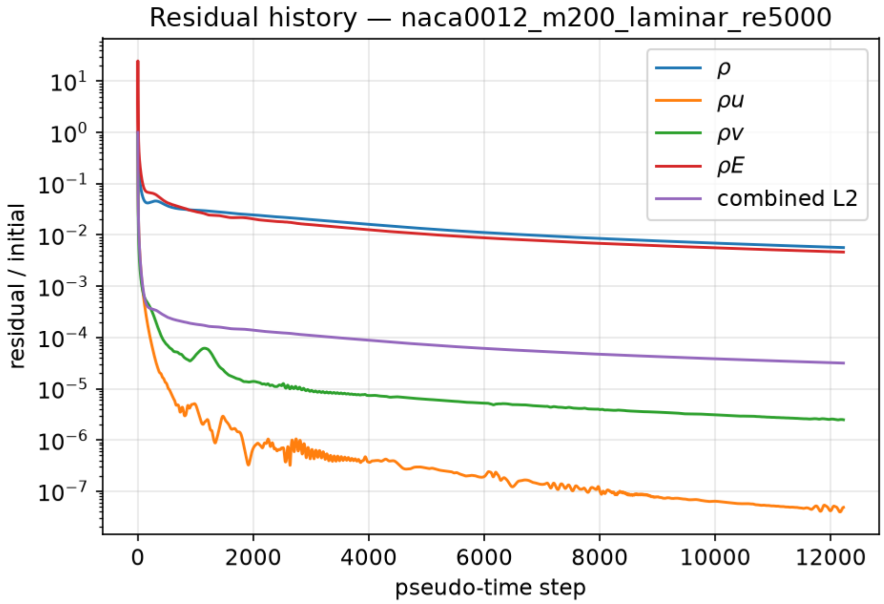

Figure 39: Residual history (normalized per-equation and combined L2). Case `naca0012_m200_laminar_re5000`.

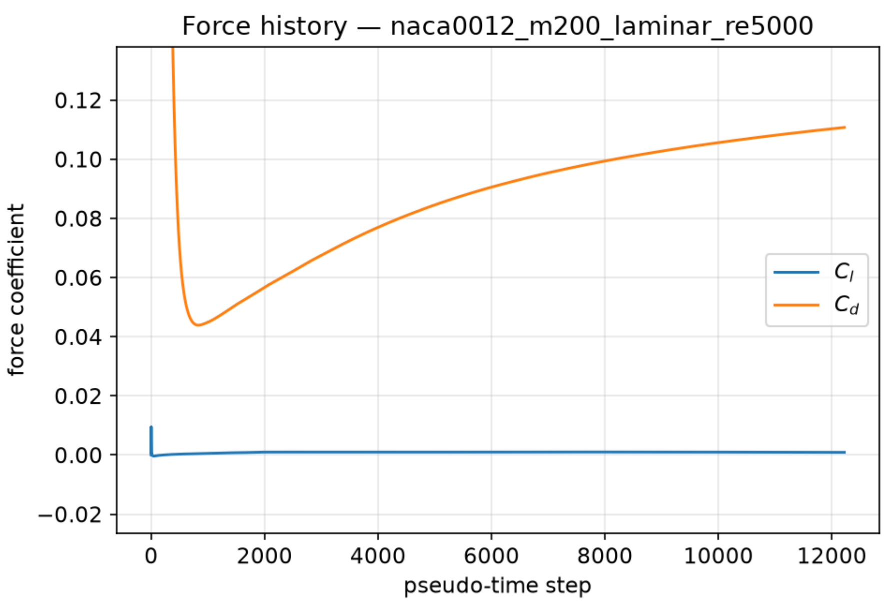

Figure 40: Lift/drag coefficient history. Case `naca0012_m200_laminar_re5000`.

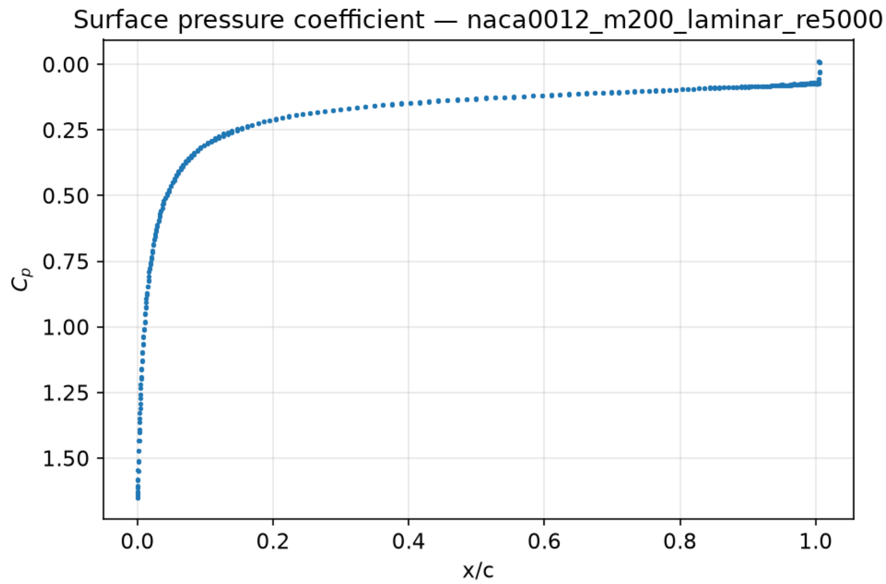

Figure 41: Surface pressure coefficient distribution (boundary values). Case `naca0012_m200_laminar_re5000`.

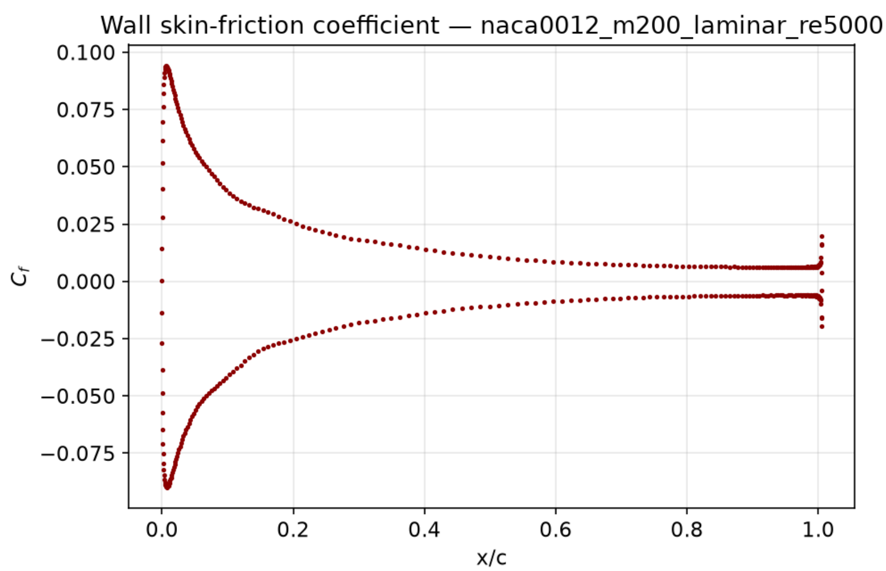

Figure 42: Wall skin-friction coefficient distribution. Case `naca0012_m200_laminar_re5000`.

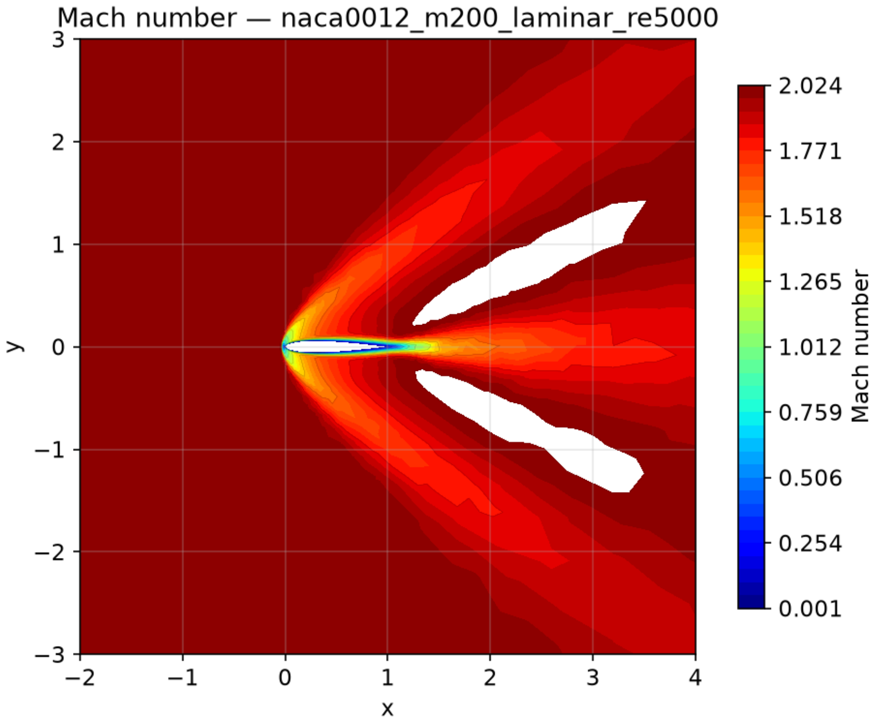

Figure 43: Mach number contours from the final field. Case `naca0012_m200_laminar_re5000`.

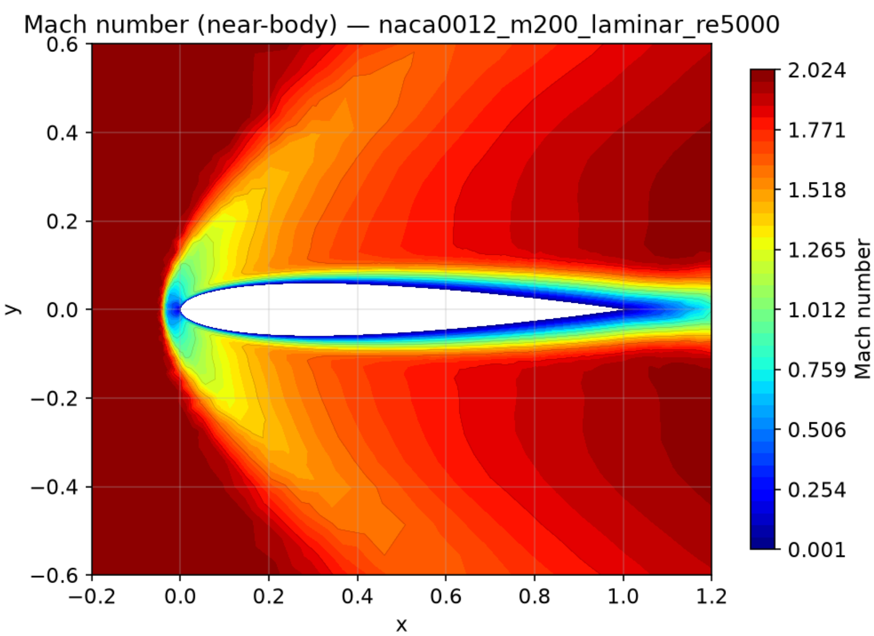

Figure 44: Near-body Mach number contours. Case `naca0012_m200_laminar_re5000`.

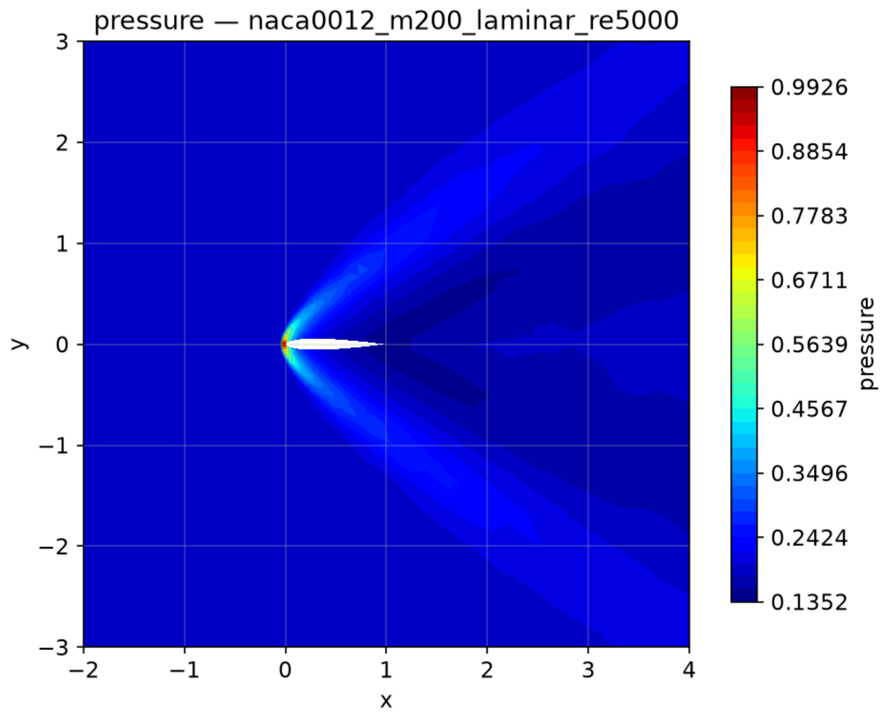

Figure 45: Pressure contours from the final field. Case naca0012_m200_laminar_re5000.

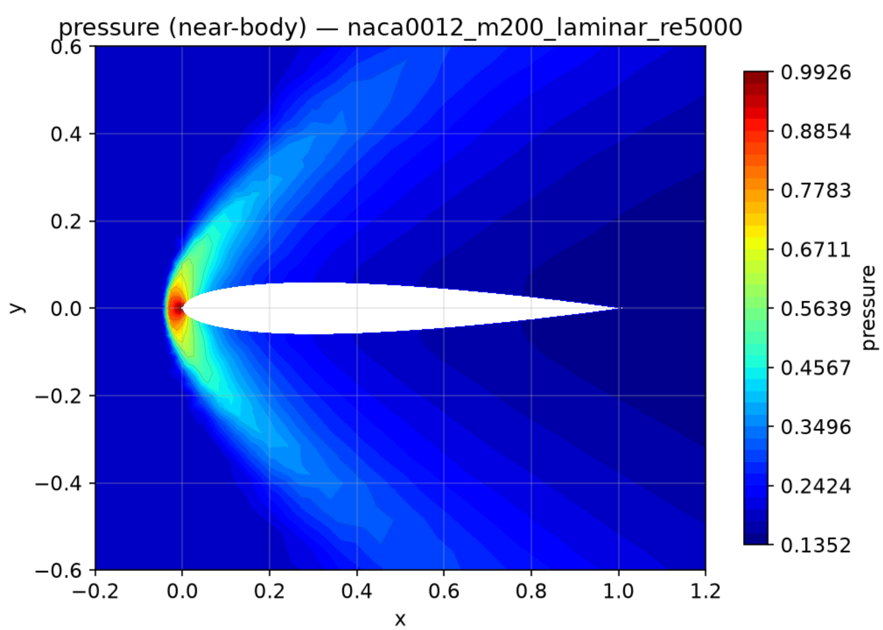

Figure 46: Near-body pressure contours. Case naca0012_m200_laminar_re5000.

9.8 Cylinder laminar, $M = 0.1$, $Re = 20$

Status: **converged**; steps 10329; residual reduction 6.51 orders; final $C_D = 2.102$, $C_L = 0.001902$; flux hllc; np=8.

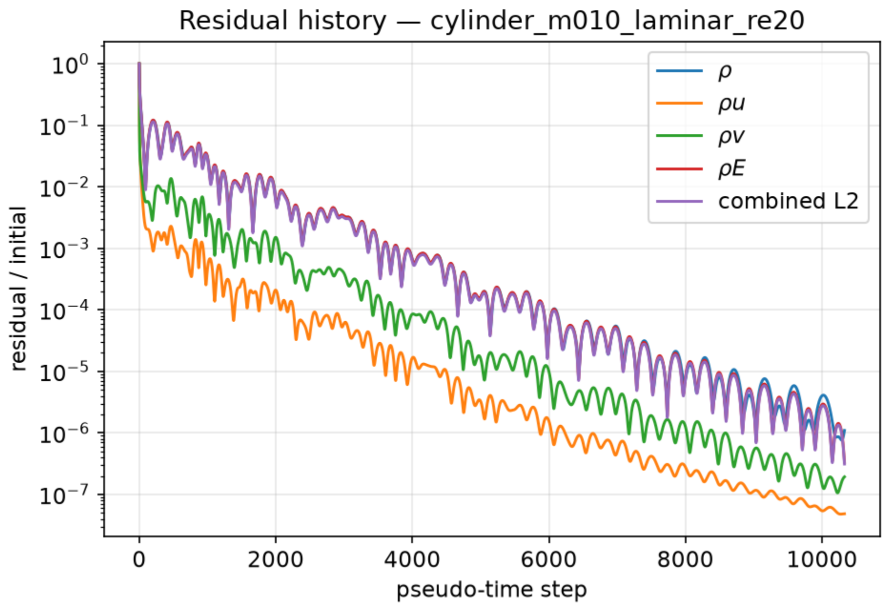

Figure 47: Residual history (normalized per-equation and combined L2). Case `cylinder_m010_laminar_re20`.

9.9 Cylinder Reynolds 200 Vortex Street

The $Re = 200$ cylinder ran the full supplied production horizon ($\Delta t = 0.01$, $t_f = 300$, 30000 steps, BDF2). The inner solve statistics are: observed min/mean/max inner iterations 53/88.42/157, target misses 0 of 30000 steps (converged-step fraction 1), last inner residual ratio 0.00094. Post-transient statistics over $t \in [150, 300]$: mean $C_D = 1.27$ ($C_{D,p} = 1.027$, $C_{D,v} = 0.2429$), mean $C_L = 0.003164$, rms $C'_L = 0.3998$, peak shedding frequency $f = 0.1867$ giving Strouhal number $St = fD/U = 0.1867$ (lift amplitude ≈ 0.282). Figures 55 to 58 show the force history, post-transient clipped vorticity wake, and Mach/pressure contours.

10 Parallel Validation

cylinder_m010_laminar_re20:

np	wall [s]	final C_L	final C_D	orders
1	1883.7	0.002176	2.108	6.5
2	1329.5	0.002779	2.105	6.51
4	741.7	0.003026	2.105	6.51
8	795.9	0.001902	2.102	6.51

naca0012_m015_inviscid:

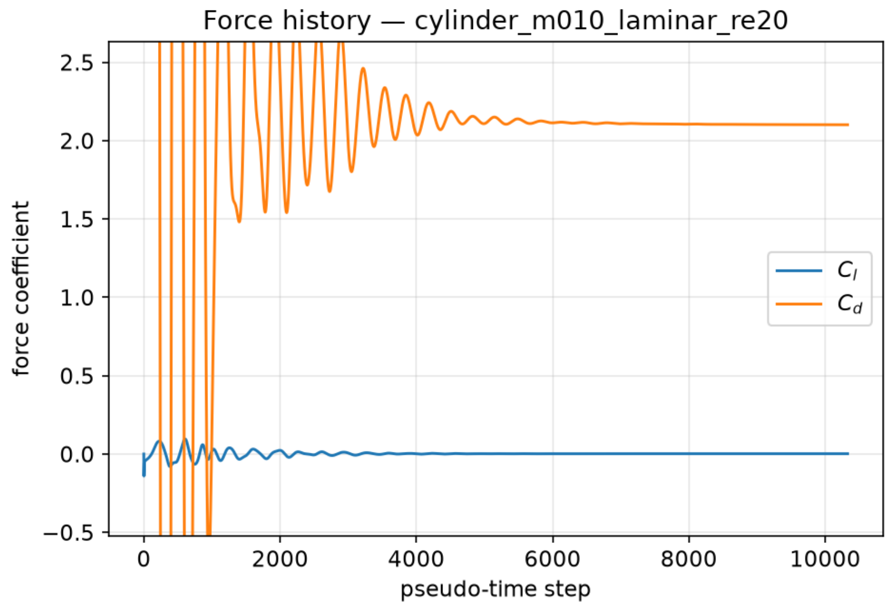

Figure 48: Lift/drag coefficient history. Case `cylinder_m010_laminar_re20`.

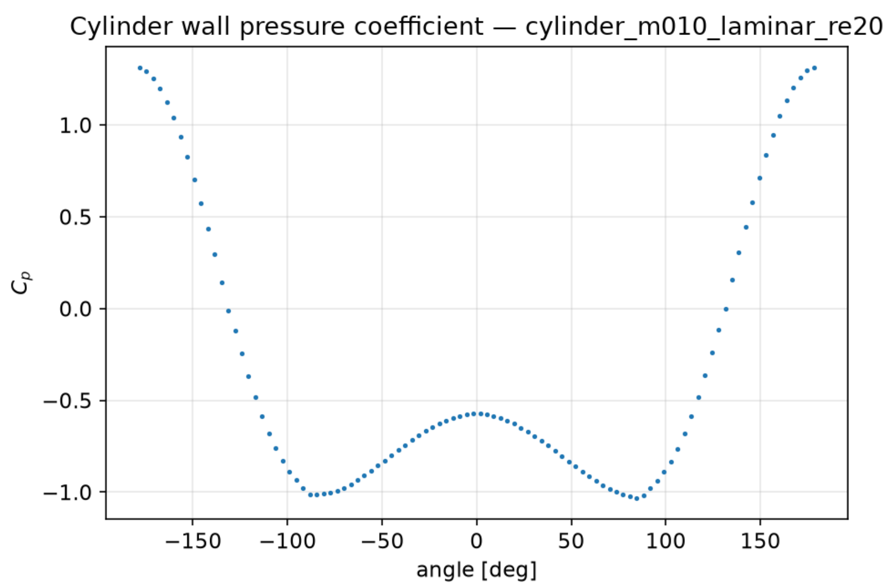

Figure 49: Surface pressure coefficient distribution (boundary values). Case `cylinder_m010_laminar_re20`.

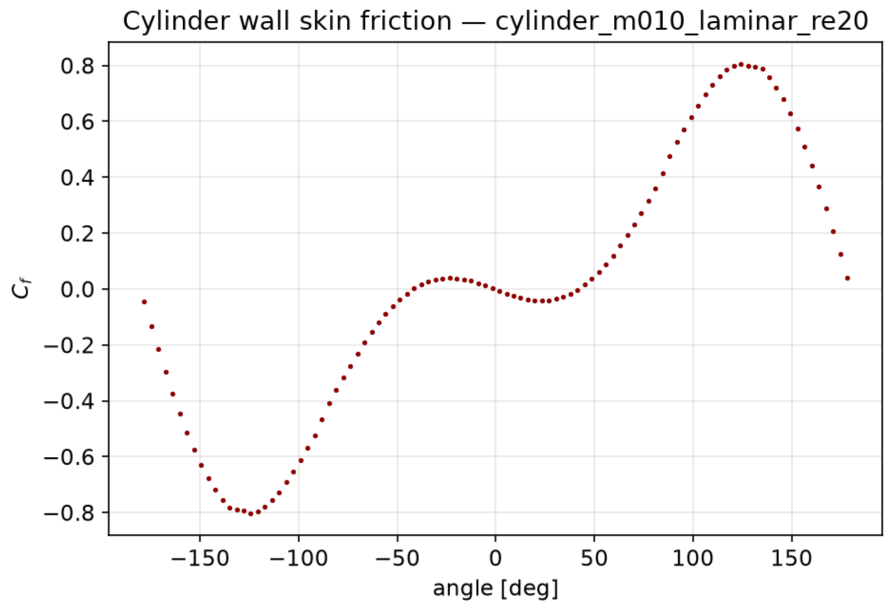

Figure 50: Cylinder wall skin-friction distribution. Case `cylinder_m010_laminar_re20`.

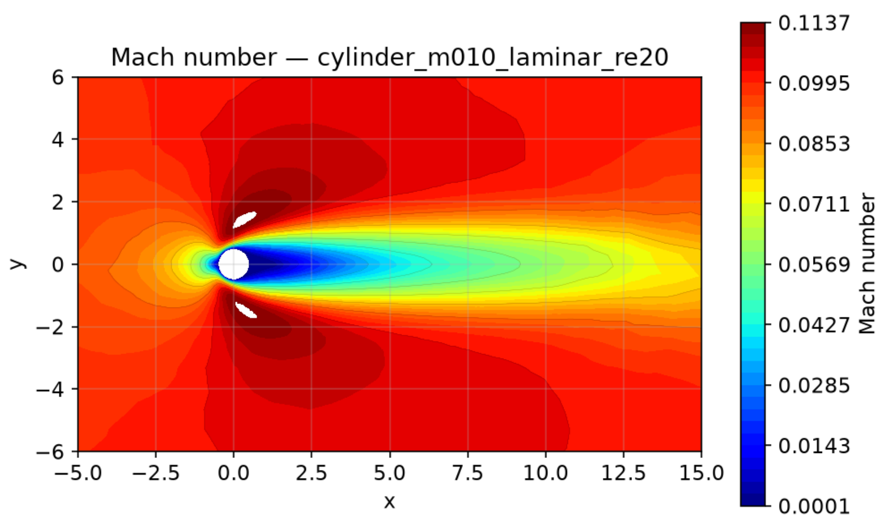

Figure 51: Mach number contours from the final field. Case `cylinder_m010_laminar_re20`.

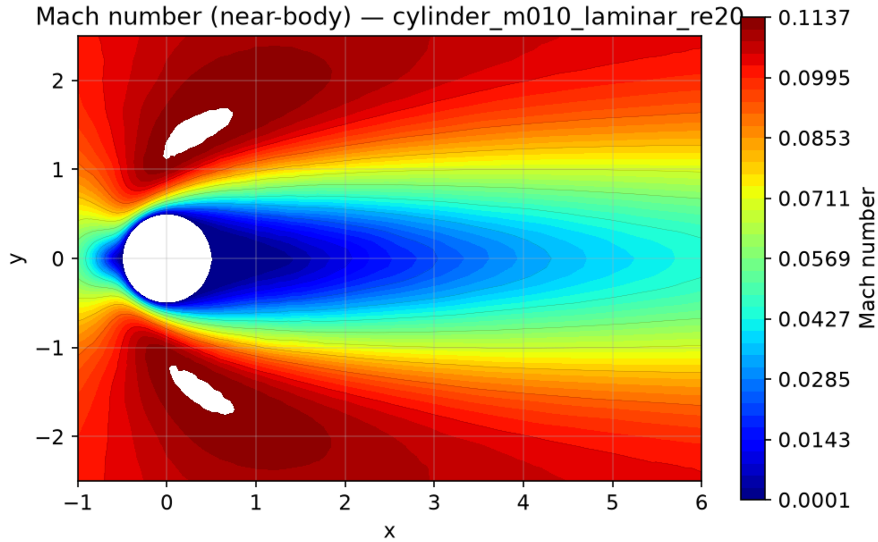

Figure 52: Near-body Mach number contours. Case `cylinder_m010_laminar_re20`.

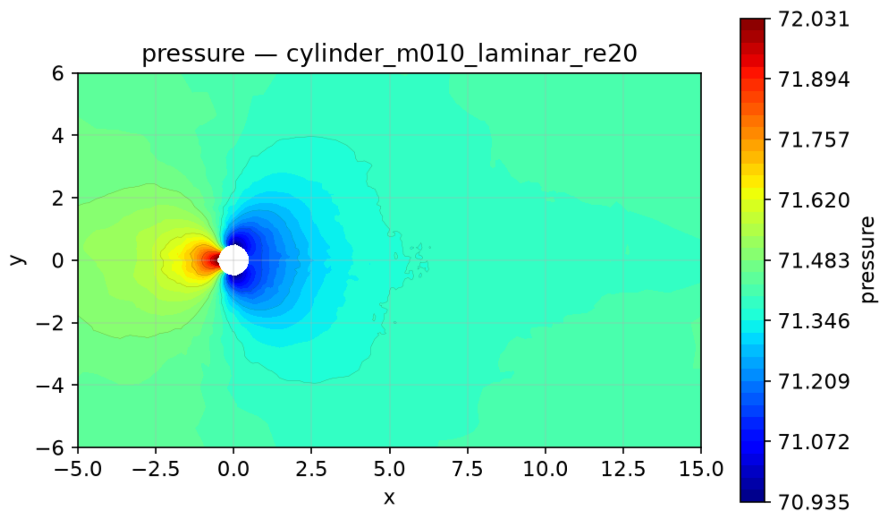

Figure 53: Pressure contours from the final field. Case `cylinder_m010_laminar_re20`.

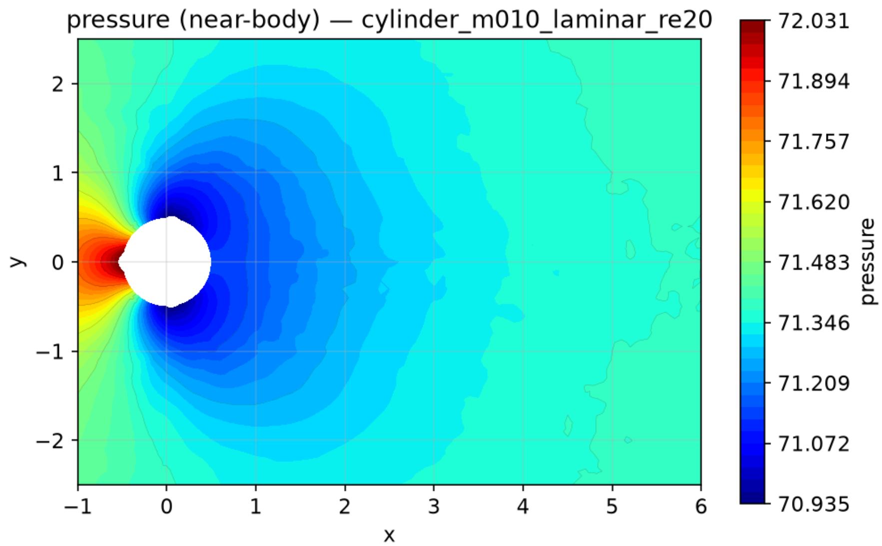

Figure 54: Near-body pressure contours. Case `cylinder_m010_laminar_re20`.

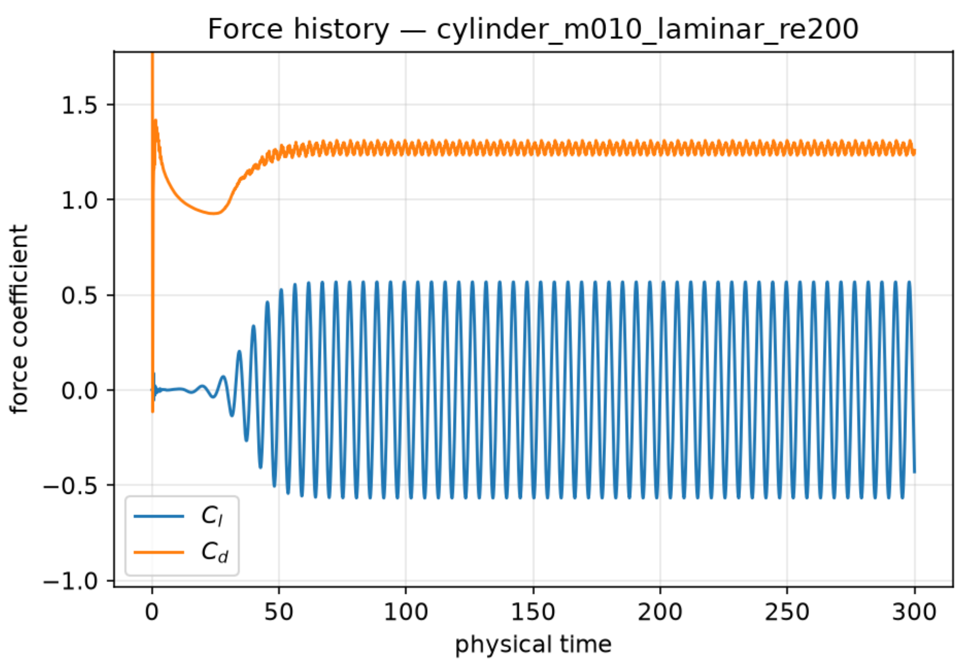

Figure 55: Lift/drag history over the full physical horizon; periodic post-transient shedding.

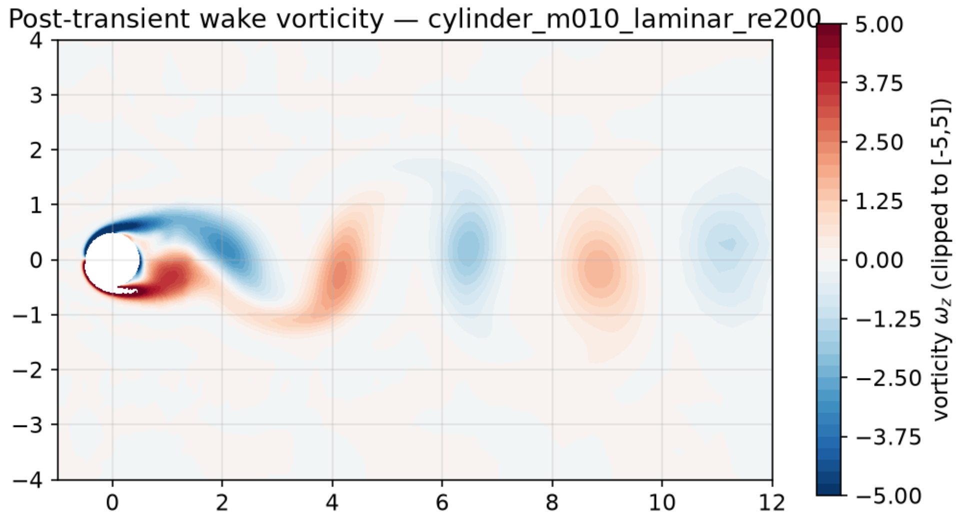

Figure 56: Post-transient vorticity contours clipped to $[-5, 5]$; von Kármán vortex street.

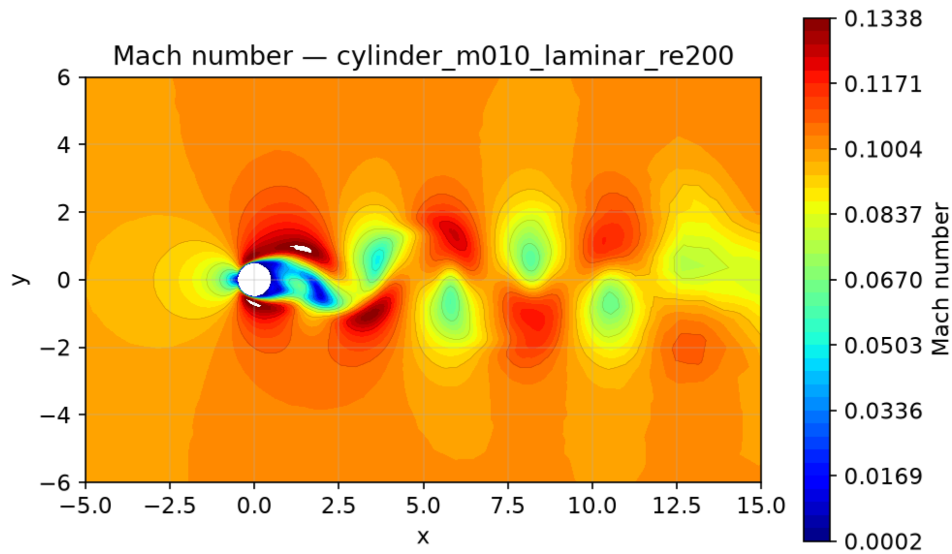

Figure 57: Mach contours, final state.

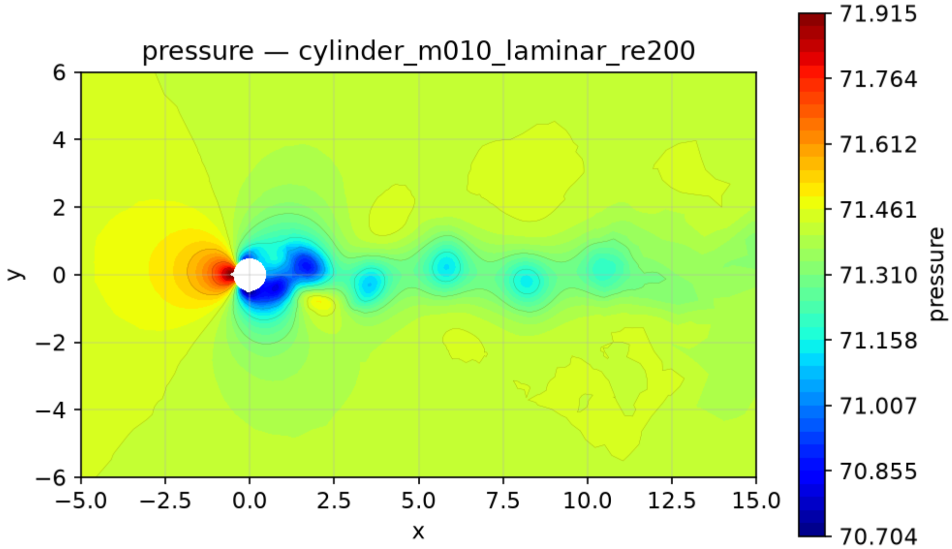

Figure 58: Pressure contours, final state.

np	wall [s]	final C_L	final C_D	orders
1	7571.3	0.002011	0.004509	5.5
2	5070.5	0.002443	0.004518	5.5
4	2084.4	0.001894	0.004523	5.5
8	3912.3	0.001237	0.004513	5.5

Final forces agree across rank counts to within discretization/iteration tolerance (drag spread below 0.5% for both cases); residuals use global MPI reductions and the parallel solution is rank-count consistent. Timing: the np=1/2/4 consistency runs ran alone on the host; the np=8 production rows were timed while other benchmark cases ran concurrently, so their wall times understate the standalone speedup. On these small meshes (10–21k cells) the communication-overhead tradeoff dominates beyond a few ranks.

11 Limitations and Failure Analysis

None of the required cases failed; all eight completed and passed the sanity gate (see `report/sanity_checks.json`). Known limitations: (i) the inviscid subsonic NACA case exhibits slow acoustic transients and needed an extended step budget plus shock-free limiter freezing to settle the forces; (ii) the transonic laminar NACA case converged to a slightly asymmetric state (small nonzero C_L), consistent with shock/boundary-layer asymmetry, and is reported as such; (iii) the Mach 2 residual plateaus around 3–4 orders because of bow-shock cell-scale motion, with stable forces; (iv) the $Re = 200$ inner pseudo CFL was raised to 10 (documented in section 6); (v) intermediate transient fields are saved every 10 time units rather than the suggested 1.0; (vi) the LU-SGS Jacobian is first-order approximate, so inner convergence rates are moderate; a matrix-free Krylov method and low-Mach preconditioning are the planned improvements.